```
"C:\Users\marci_plgx30x\Desktop\Arquivos\Projetos do Mestrado (Git)\Projeto-Classificadores\.cnn_venv\Scripts\python.exe" "C:\Users\marci_plgx30x\Desktop\Arquivos\Projetos do Mestrado (Git)\Projeto-Classificadores\Project_CNN\DenseNet\DenseNet_Main.py"
2026-05-22 19:09:48.062732: I tensorflow/core/platform/cpu_feature_guard.cc:193] This TensorFlow binary is optimized with oneAPI Deep Neural Network Library (oneDNN) to use the following CPU instructions in performance-critical operations:  AVX  AVX2
To enable them in other operations, rebuild TensorFlow with the appropriate compiler flags.
2026-05-22 19:09:48.119391: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1616] Created device  /job:localhost/replica:0/task:0/device:GPU:0 with 3539 MB memory:  -> device: 0, name: NVIDIA GeForce RTX 4050 Laptop GPU, pci bus id: 0000:01:00.0, compute capability: 8.9
Mixed precision ativada (float16) para acelerar o treinamento.
GPU detectada: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
Seeds fixados (seed=42) para reprodutibilidade.


------------------------------------------------------------
                                    INICIANDO PIPELINE DE EXECUÇÃO

------------------------------------------------------------
Found 11074 files belonging to 3 classes.
Found 2767 files belonging to 3 classes.

>> LOGS PROVENIENTES DO CARREGAMENTO DO DATASET
Treinamento:  11074
```

```
Validação:  2767
Índices:  ['Indolor', 'Muita dor', 'Pouca dor']
Classes detectadas: 3

Model: "model"
__________________________________________________
_____________________________________________
 Layer (type)                    Output Shape
Param #     Connected to
==================================================
==============================================
 input_1 (InputLayer)            [(None, 128, 128, 3
0           []
                                 )]


 conv2d (Conv2D)                 (None, 128, 128, 48
1296        ['input_1[0][0]']
                                 )


 batch_normalization (BatchNorm  (None, 128, 128, 48
 192        ['conv2d[0][0]']
 alization
)                    )


 activation (Activation)         (None, 128, 128, 48
0           ['batch_normalization[0][0]']
                                 )


 batch_normalization_1 (BatchNo  (None, 128, 128, 48
 192        ['activation[0][0]']
 rmalization
)                   )
```

```
40
41  activation_1 (Activation)      (None, 128, 128, 48
   0           ['batch_normalization_1[0][0]']
42                                 )
43
44  conv2d_1 (Conv2D)              (None, 128, 128, 96
   4608        ['activation_1[0][0]']
45                                 )
46
47  batch_normalization_2 (BatchNo  (None, 128, 128, 96
     384        ['conv2d_1[0][0]']
48  rmalization
   )                   )
49
50  activation_2 (Activation)      (None, 128, 128, 96
   0           ['batch_normalization_2[0][0]']
51                                 )
52
53  conv2d_2 (Conv2D)              (None, 128, 128, 24
   20736       ['activation_2[0][0]']
54                                 )
55
56  concatenate (Concatenate)      (None, 128, 128, 72
   0           ['activation[0][0]',
57                                 )
               'conv2d_2[0][0]']
58
59  batch_normalization_3 (BatchNo  (None, 128, 128, 72
     288        ['concatenate[0][0]']
```

```
60  rmalization
   )                   )

61

62  activation_3 (Activation)      (None, 128, 128, 72
     0           ['batch_normalization_3[0][0]']
63                                  )

64

65  conv2d_3 (Conv2D)              (None, 128, 128, 96
     6912        ['activation_3[0][0]']
66                                  )

67

68  batch_normalization_4 (BatchNo  (None, 128, 128, 96
     384        ['conv2d_3[0][0]']
69  rmalization
   )                   )

70

71  activation_4 (Activation)      (None, 128, 128, 96
     0           ['batch_normalization_4[0][0]']
72                                  )

73

74  conv2d_4 (Conv2D)              (None, 128, 128, 24
     20736       ['activation_4[0][0]']
75                                  )

76

77  concatenate_1 (Concatenate)    (None, 128, 128, 96
     0           ['concatenate[0][0]',
78                                  )
                 'conv2d_4[0][0]']
79
```

```
79
80  batch_normalization_5 (BatchNo  (None, 128, 128, 96
     384         ['concatenate_1[0][0]']
81  rmalization
   )                  )

82

83  activation_5 (Activation)      (None, 128, 128, 96
     0           ['batch_normalization_5[0][0]']
84                                 )

85

86  conv2d_5 (Conv2D)              (None, 128, 128, 96
     9216        ['activation_5[0][0]']
87                                 )

88

89  batch_normalization_6 (BatchNo  (None, 128, 128, 96
     384         ['conv2d_5[0][0]']
90  rmalization
   )                  )

91

92  activation_6 (Activation)      (None, 128, 128, 96
     0           ['batch_normalization_6[0][0]']
93                                 )

94

95  conv2d_6 (Conv2D)              (None, 128, 128, 24
     20736       ['activation_6[0][0]']
96                                 )

97

98  concatenate_2 (Concatenate)    (None, 128, 128, 12
     0           ['concatenate_1[0][0]',
```

```
 99                                  0
    )                                 'conv2d_6[0][0
    ]']
100

101  batch_normalization_7 (BatchNo  (None, 128, 128, 12
      480        ['concatenate_2[0][0]']
102  rmalization)                    0
    )

103

104  activation_7 (Activation)       (None, 128, 128, 12
      0          ['batch_normalization_7[0][0]']
105                                  0
    )

106

107  conv2d_7 (Conv2D)               (None, 128, 128, 96
      11520      ['activation_7[0][0]']
108                                  )

109

110  batch_normalization_8 (BatchNo  (None, 128, 128, 96
      384        ['conv2d_7[0][0]']
111  rmalization
    )                   )

112

113  activation_8 (Activation)       (None, 128, 128, 96
      0          ['batch_normalization_8[0][0]']
114                                  )

115

116  conv2d_8 (Conv2D)               (None, 128, 128, 24
      20736      ['activation_8[0][0]']
117                                  )
```

```
117
118
119  concatenate_3 (Concatenate)    (None, 128, 128, 14
      0           ['concatenate_2[0][0]',
120                                 4
    )                                'conv2d_8[0][0
    ]']
121
122  batch_normalization_9 (BatchNo  (None, 128, 128, 14
      576        ['concatenate_3[0][0]']
123  rmalization)                   4
    )
124
125  activation_9 (Activation)      (None, 128, 128, 14
      0           ['batch_normalization_9[0][0]']
126                                 4
    )
127
128  conv2d_9 (Conv2D)              (None, 128, 128, 96
      13824       ['activation_9[0][0]']
129                                 )
130
131  batch_normalization_10 (BatchN  (None, 128, 128, 96
      384        ['conv2d_9[0][0]']
132  ormalization
    )                 )
133
134  activation_10 (Activation)     (None, 128, 128, 96
      0           ['batch_normalization_10[0][0]']
135                                 )
```

```
136
137  conv2d_10 (Conv2D)              (None, 128, 128, 24
      20736       ['activation_10[0][0]']
138                                  )
139
140  concatenate_4 (Concatenate)     (None, 128, 128, 16
      0           ['concatenate_3[0][0]',
141                                  8
    )                                 'conv2d_10[0][0
    ]']
142
143  batch_normalization_11 (BatchN  (None, 128, 128, 16
      672         ['concatenate_4[0][0]']
144  ormalization)                   8
    )
145
146  activation_11 (Activation)      (None, 128, 128, 16
      0           ['batch_normalization_11[0][0]']
147                                  8
    )
148
149  conv2d_11 (Conv2D)              (None, 128, 128, 96
      16128       ['activation_11[0][0]']
150                                  )
151
152  batch_normalization_12 (BatchN  (None, 128, 128, 96
      384         ['conv2d_11[0][0]']
153  ormalization
    )                 )
154
```

```
154
155  activation_12 (Activation)     (None, 128, 128, 96
      0           ['batch_normalization_12[0][0]']
156                                 )

157

158  conv2d_12 (Conv2D)             (None, 128, 128, 24
      20736       ['activation_12[0][0]']
159                                 )

160

161  concatenate_5 (Concatenate)    (None, 128, 128, 19
      0           ['concatenate_4[0][0]',
162                                 2
    )                                'conv2d_12[0][0
    ]']
163

164  batch_normalization_13 (BatchN  (None, 128, 128, 19
      768        ['concatenate_5[0][0]']
165  ormalization)                  2
    )

166

167  activation_13 (Activation)     (None, 128, 128, 19
      0           ['batch_normalization_13[0][0]']
168                                 2
    )

169

170  conv2d_13 (Conv2D)             (None, 128, 128, 96
      18432       ['activation_13[0][0]']
171                                 )

172

173  average_pooling2d (AveragePool  (None, 64, 64, 96
```

```
173 )  0           ['conv2d_13[0][0]']
174  ing2D
    )

175

176  batch_normalization_14 (BatchN  (None, 64, 64, 96
    )  384         ['average_pooling2d[0][0]']
177  ormalization
    )

178

179  activation_14 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_14[0][0]']
180

181  conv2d_14 (Conv2D)             (None, 64, 64, 96
    )   9216        ['activation_14[0][0]']
182

183  batch_normalization_15 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_14[0][0]']
184  ormalization
    )

185

186  activation_15 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_15[0][0]']
187

188  conv2d_15 (Conv2D)             (None, 64, 64, 24
    )   20736       ['activation_15[0][0]']
189

190  concatenate_6 (Concatenate)    (None, 64, 64, 120
    )  0           ['average_pooling2d[0][0]',
191
                 'conv2d_15[0][0]']
192
```

```
192
193  batch_normalization_16 (BatchN  (None, 64, 64, 120
    )  480         ['concatenate_6[0][0]']
194  ormalization
    )

195

196  activation_16 (Activation)     (None, 64, 64, 120
    )  0           ['batch_normalization_16[0][0]']
197

198  conv2d_16 (Conv2D)             (None, 64, 64, 96
    )   11520       ['activation_16[0][0]']
199

200  batch_normalization_17 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_16[0][0]']
201  ormalization
    )

202

203  activation_17 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_17[0][0]']
204

205  conv2d_17 (Conv2D)             (None, 64, 64, 24
    )   20736       ['activation_17[0][0]']
206

207  concatenate_7 (Concatenate)    (None, 64, 64, 144
    )  0           ['concatenate_6[0][0]',
208
                    'conv2d_17[0][0]']
209

210  batch_normalization_18 (BatchN  (None, 64, 64, 144
    )  576         ['concatenate_7[0][0]']
211  ormalization
    )
```

```
211
212

213  activation_18 (Activation)     (None, 64, 64, 144
    )  0           ['batch_normalization_18[0][0]']
214

215  conv2d_18 (Conv2D)             (None, 64, 64, 96
    )   13824       ['activation_18[0][0]']
216

217  batch_normalization_19 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_18[0][0]']
218  ormalization
    )

219

220  activation_19 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_19[0][0]']
221

222  conv2d_19 (Conv2D)             (None, 64, 64, 24
    )   20736       ['activation_19[0][0]']
223

224  concatenate_8 (Concatenate)    (None, 64, 64, 168
    )  0           ['concatenate_7[0][0]',
225
                   'conv2d_19[0][0]']
226

227  batch_normalization_20 (BatchN  (None, 64, 64, 168
    )  672        ['concatenate_8[0][0]']
228  ormalization
    )

229

230  activation_20 (Activation)     (None, 64, 64, 168
    )  0           ['batch_normalization_20[0][0]']
```

```
231
232  conv2d_20 (Conv2D)              (None, 64, 64, 96
    )   16128       ['activation_20[0][0]']
233
234  batch_normalization_21 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_20[0][0]']
235  ormalization
    )
236
237  activation_21 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_21[0][0]']
238
239  conv2d_21 (Conv2D)              (None, 64, 64, 24
    )   20736       ['activation_21[0][0]']
240
241  concatenate_9 (Concatenate)    (None, 64, 64, 192
    )  0           ['concatenate_8[0][0]',
242
                   'conv2d_21[0][0]']
243
244  batch_normalization_22 (BatchN  (None, 64, 64, 192
    )  768        ['concatenate_9[0][0]']
245  ormalization
    )
246
247  activation_22 (Activation)     (None, 64, 64, 192
    )  0           ['batch_normalization_22[0][0]']
248
249  conv2d_22 (Conv2D)              (None, 64, 64, 96
    )   18432       ['activation_22[0][0]']
250
```

```
250
251  batch_normalization_23 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_22[0][0]']
252  ormalization
    )

253

254  activation_23 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_23[0][0]']
255

256  conv2d_23 (Conv2D)             (None, 64, 64, 24
    )   20736       ['activation_23[0][0]']
257

258  concatenate_10 (Concatenate)   (None, 64, 64, 216
    )  0           ['concatenate_9[0][0]',
259
                    'conv2d_23[0][0]']
260

261  batch_normalization_24 (BatchN  (None, 64, 64, 216
    )  864         ['concatenate_10[0][0]']
262  ormalization
    )

263

264  activation_24 (Activation)     (None, 64, 64, 216
    )  0           ['batch_normalization_24[0][0]']
265

266  conv2d_24 (Conv2D)             (None, 64, 64, 96
    )   20736       ['activation_24[0][0]']
267

268  batch_normalization_25 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_24[0][0]']
269  ormalization
    )
```

```
269
270
271  activation_25 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_25[0][0]']
272
273  conv2d_25 (Conv2D)             (None, 64, 64, 24
    )   20736       ['activation_25[0][0]']
274
275  concatenate_11 (Concatenate)   (None, 64, 64, 240
    )  0           ['concatenate_10[0][0]',
276
                   'conv2d_25[0][0]']
277
278  batch_normalization_26 (BatchN  (None, 64, 64, 240
    )  960         ['concatenate_11[0][0]']
279  ormalization
    )
280
281  activation_26 (Activation)     (None, 64, 64, 240
    )  0           ['batch_normalization_26[0][0]']
282
283  conv2d_26 (Conv2D)             (None, 64, 64, 96
    )   23040       ['activation_26[0][0]']
284
285  batch_normalization_27 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_26[0][0]']
286  ormalization
    )
287
288  activation_27 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_27[0][0]']
```

```
289
290  conv2d_27 (Conv2D)              (None, 64, 64, 24
    )   20736       ['activation_27[0][0]']
291
292  concatenate_12 (Concatenate)   (None, 64, 64, 264
    )  0           ['concatenate_11[0][0]',
293
                    'conv2d_27[0][0]']
294
295  batch_normalization_28 (BatchN  (None, 64, 64, 264
    )  1056       ['concatenate_12[0][0]']
296  ormalization
    )
297
298  activation_28 (Activation)     (None, 64, 64, 264
    )  0           ['batch_normalization_28[0][0]']
299
300  conv2d_28 (Conv2D)              (None, 64, 64, 96
    )   25344       ['activation_28[0][0]']
301
302  batch_normalization_29 (BatchN  (None, 64, 64, 96
    )  384         ['conv2d_28[0][0]']
303  ormalization
    )
304
305  activation_29 (Activation)     (None, 64, 64, 96
    )   0           ['batch_normalization_29[0][0]']
306
307  conv2d_29 (Conv2D)              (None, 64, 64, 24
    )   20736       ['activation_29[0][0]']
308
```

```
308
309  concatenate_13 (Concatenate)   (None, 64, 64, 288
    )  0           ['concatenate_12[0][0]',
310
                    'conv2d_29[0][0]']
311

312  batch_normalization_30 (BatchN  (None, 64, 64, 288
    )  1152        ['concatenate_13[0][0]']
313  ormalization
    )

314

315  activation_30 (Activation)     (None, 64, 64, 288
    )  0           ['batch_normalization_30[0][0]']
316

317  conv2d_30 (Conv2D)             (None, 64, 64, 144
    )  41472       ['activation_30[0][0]']
318

319  average_pooling2d_1 (AveragePo  (None, 32, 32, 144
    )  0           ['conv2d_30[0][0]']
320  oling2D
    )

321

322  batch_normalization_31 (BatchN  (None, 32, 32, 144
    )  576         ['average_pooling2d_1[0][0]']
323  ormalization
    )

324

325  activation_31 (Activation)     (None, 32, 32, 144
    )  0           ['batch_normalization_31[0][0]']
326

327  conv2d_31 (Conv2D)             (None, 32, 32, 96
```

```
327 )   13824       ['activation_31[0][0]']
328
329  batch_normalization_32 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_31[0][0]']
330  ormalization
    )
331
332  activation_32 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_32[0][0]']
333
334  conv2d_32 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_32[0][0]']
335
336  concatenate_14 (Concatenate)   (None, 32, 32, 168
    )  0           ['average_pooling2d_1[0][0]',
337
                   'conv2d_32[0][0]']
338
339  batch_normalization_33 (BatchN  (None, 32, 32, 168
    )  672        ['concatenate_14[0][0]']
340  ormalization
    )
341
342  activation_33 (Activation)     (None, 32, 32, 168
    )  0           ['batch_normalization_33[0][0]']
343
344  conv2d_33 (Conv2D)             (None, 32, 32, 96
    )   16128       ['activation_33[0][0]']
345
346  batch_normalization_34 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_33[0][0]']
```

```
347  ormalization
    )

348

349  activation_34 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_34[0][0]']
350

351  conv2d_34 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_34[0][0]']
352

353  concatenate_15 (Concatenate)   (None, 32, 32, 192
    )  0           ['concatenate_14[0][0]',
354
                   'conv2d_34[0][0]']
355

356  batch_normalization_35 (BatchN  (None, 32, 32, 192
    )  768         ['concatenate_15[0][0]']
357  ormalization
    )

358

359  activation_35 (Activation)     (None, 32, 32, 192
    )  0           ['batch_normalization_35[0][0]']
360

361  conv2d_35 (Conv2D)             (None, 32, 32, 96
    )   18432       ['activation_35[0][0]']
362

363  batch_normalization_36 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_35[0][0]']
364  ormalization
    )

365
```

```
366  activation_36 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_36[0][0]']
367
368  conv2d_36 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_36[0][0]']
369
370  concatenate_16 (Concatenate)   (None, 32, 32, 216
    )  0           ['concatenate_15[0][0]',
371
                   'conv2d_36[0][0]']
372
373  batch_normalization_37 (BatchN  (None, 32, 32, 216
    )  864         ['concatenate_16[0][0]']
374  ormalization
    )
375
376  activation_37 (Activation)     (None, 32, 32, 216
    )  0           ['batch_normalization_37[0][0]']
377
378  conv2d_37 (Conv2D)             (None, 32, 32, 96
    )   20736       ['activation_37[0][0]']
379
380  batch_normalization_38 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_37[0][0]']
381  ormalization
    )
382
383  activation_38 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_38[0][0]']
384
385  conv2d_38 (Conv2D)             (None, 32, 32, 24
```

```
385 )   20736       ['activation_38[0][0]']
386
387  concatenate_17 (Concatenate)   (None, 32, 32, 240
    )  0           ['concatenate_16[0][0]',
388
                    'conv2d_38[0][0]']
389
390  batch_normalization_39 (BatchN  (None, 32, 32, 240
    )  960         ['concatenate_17[0][0]']
391  ormalization
    )
392
393  activation_39 (Activation)     (None, 32, 32, 240
    )  0           ['batch_normalization_39[0][0]']
394
395  conv2d_39 (Conv2D)             (None, 32, 32, 96
    )   23040       ['activation_39[0][0]']
396
397  batch_normalization_40 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_39[0][0]']
398  ormalization
    )
399
400  activation_40 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_40[0][0]']
401
402  conv2d_40 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_40[0][0]']
403
404  concatenate_18 (Concatenate)   (None, 32, 32, 264
    )  0           ['concatenate_17[0][0]',
```

```
405
                'conv2d_40[0][0]']
406

407  batch_normalization_41 (BatchN  (None, 32, 32, 264
    )  1056       ['concatenate_18[0][0]']
408  ormalization
    )

409

410  activation_41 (Activation)     (None, 32, 32, 264
    )  0           ['batch_normalization_41[0][0]']
411

412  conv2d_41 (Conv2D)             (None, 32, 32, 96
    )   25344       ['activation_41[0][0]']
413

414  batch_normalization_42 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_41[0][0]']
415  ormalization
    )

416

417  activation_42 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_42[0][0]']
418

419  conv2d_42 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_42[0][0]']
420

421  concatenate_19 (Concatenate)   (None, 32, 32, 288
    )  0           ['concatenate_18[0][0]',
422
                'conv2d_42[0][0]']
423

424  batch_normalization_43 (BatchN  (None, 32, 32, 288
```

```
424 )  1152       ['concatenate_19[0][0]']
425  ormalization
    )

426

427  activation_43 (Activation)     (None, 32, 32, 288
    )  0           ['batch_normalization_43[0][0]']
428

429  conv2d_43 (Conv2D)             (None, 32, 32, 96
    )   27648       ['activation_43[0][0]']
430

431  batch_normalization_44 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_43[0][0]']
432  ormalization
    )

433

434  activation_44 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_44[0][0]']
435

436  conv2d_44 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_44[0][0]']
437

438  concatenate_20 (Concatenate)   (None, 32, 32, 312
    )  0           ['concatenate_19[0][0]',
439
                   'conv2d_44[0][0]']
440

441  batch_normalization_45 (BatchN  (None, 32, 32, 312
    )  1248       ['concatenate_20[0][0]']
442  ormalization
    )

443
```

```
443
444  activation_45 (Activation)     (None, 32, 32, 312
    )  0           ['batch_normalization_45[0][0]']
445
446  conv2d_45 (Conv2D)             (None, 32, 32, 96
    )   29952       ['activation_45[0][0]']
447
448  batch_normalization_46 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_45[0][0]']
449  ormalization
    )
450
451  activation_46 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_46[0][0]']
452
453  conv2d_46 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_46[0][0]']
454
455  concatenate_21 (Concatenate)   (None, 32, 32, 336
    )  0           ['concatenate_20[0][0]',
456
                   'conv2d_46[0][0]']
457
458  batch_normalization_47 (BatchN  (None, 32, 32, 336
    )  1344       ['concatenate_21[0][0]']
459  ormalization
    )
460
461  activation_47 (Activation)     (None, 32, 32, 336
    )  0           ['batch_normalization_47[0][0]']
462
```

```
463  conv2d_47 (Conv2D)              (None, 32, 32, 96
    )   32256       ['activation_47[0][0]']
464
465  batch_normalization_48 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_47[0][0]']
466  ormalization
    )
467
468  activation_48 (Activation)      (None, 32, 32, 96
    )   0           ['batch_normalization_48[0][0]']
469
470  conv2d_48 (Conv2D)              (None, 32, 32, 24
    )   20736       ['activation_48[0][0]']
471
472  concatenate_22 (Concatenate)    (None, 32, 32, 360
    )  0           ['concatenate_21[0][0]',
473
                    'conv2d_48[0][0]']
474
475  batch_normalization_49 (BatchN  (None, 32, 32, 360
    )  1440        ['concatenate_22[0][0]']
476  ormalization
    )
477
478  activation_49 (Activation)      (None, 32, 32, 360
    )  0           ['batch_normalization_49[0][0]']
479
480  conv2d_49 (Conv2D)              (None, 32, 32, 96
    )   34560       ['activation_49[0][0]']
481
482  batch_normalization_50 (BatchN  (None, 32, 32, 96
```

```
)  384         ['conv2d_49[0][0]']
 ormalization
)


 activation_50 (Activation)     (None, 32, 32, 96
)   0           ['batch_normalization_50[0][0]']


 conv2d_50 (Conv2D)             (None, 32, 32, 24
)   20736       ['activation_50[0][0]']


 concatenate_23 (Concatenate)   (None, 32, 32, 384
)  0           ['concatenate_22[0][0]',

                'conv2d_50[0][0]']


 batch_normalization_51 (BatchN  (None, 32, 32, 384
)  1536       ['concatenate_23[0][0]']
 ormalization
)


 activation_51 (Activation)     (None, 32, 32, 384
)  0           ['batch_normalization_51[0][0]']


 conv2d_51 (Conv2D)             (None, 32, 32, 96
)   36864       ['activation_51[0][0]']


 batch_normalization_52 (BatchN  (None, 32, 32, 96
)  384         ['conv2d_51[0][0]']
 ormalization
)


```

```
501
502  activation_52 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_52[0][0]']
503
504  conv2d_52 (Conv2D)             (None, 32, 32, 24
    )   20736       ['activation_52[0][0]']
505
506  concatenate_24 (Concatenate)   (None, 32, 32, 408
    )  0           ['concatenate_23[0][0]',
507
                    'conv2d_52[0][0]']
508
509  batch_normalization_53 (BatchN  (None, 32, 32, 408
    )  1632        ['concatenate_24[0][0]']
510  ormalization
    )
511
512  activation_53 (Activation)     (None, 32, 32, 408
    )  0           ['batch_normalization_53[0][0]']
513
514  conv2d_53 (Conv2D)             (None, 32, 32, 96
    )   39168       ['activation_53[0][0]']
515
516  batch_normalization_54 (BatchN  (None, 32, 32, 96
    )  384         ['conv2d_53[0][0]']
517  ormalization
    )
518
519  activation_54 (Activation)     (None, 32, 32, 96
    )   0           ['batch_normalization_54[0][0]']
520
```

```
521  conv2d_54 (Conv2D)              (None, 32, 32, 24
    )   20736       ['activation_54[0][0]']
522
523  concatenate_25 (Concatenate)    (None, 32, 32, 432
    )  0           ['concatenate_24[0][0]',
524
                    'conv2d_54[0][0]']
525
526  batch_normalization_55 (BatchN  (None, 32, 32, 432
    )  1728        ['concatenate_25[0][0]']
527  ormalization
    )
528
529  activation_55 (Activation)      (None, 32, 32, 432
    )  0           ['batch_normalization_55[0][0]']
530
531  conv2d_55 (Conv2D)              (None, 32, 32, 216
    )  93312       ['activation_55[0][0]']
532
533  average_pooling2d_2 (AveragePo  (None, 16, 16, 216
    )  0           ['conv2d_55[0][0]']
534  oling2D
    )
535
536  batch_normalization_56 (BatchN  (None, 16, 16, 216
    )  864         ['average_pooling2d_2[0][0]']
537  ormalization
    )
538
539  activation_56 (Activation)      (None, 16, 16, 216
    )  0           ['batch_normalization_56[0][0]']
```

```
540
541  conv2d_56 (Conv2D)              (None, 16, 16, 96
    )   20736       ['activation_56[0][0]']
542
543  batch_normalization_57 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_56[0][0]']
544  ormalization
    )
545
546  activation_57 (Activation)      (None, 16, 16, 96
    )   0           ['batch_normalization_57[0][0]']
547
548  conv2d_57 (Conv2D)              (None, 16, 16, 24
    )   20736       ['activation_57[0][0]']
549
550  concatenate_26 (Concatenate)    (None, 16, 16, 240
    )  0           ['average_pooling2d_2[0][0]',
551
                    'conv2d_57[0][0]']
552
553  batch_normalization_58 (BatchN  (None, 16, 16, 240
    )  960        ['concatenate_26[0][0]']
554  ormalization
    )
555
556  activation_58 (Activation)      (None, 16, 16, 240
    )  0           ['batch_normalization_58[0][0]']
557
558  conv2d_58 (Conv2D)              (None, 16, 16, 96
    )   23040       ['activation_58[0][0]']
559
```

```
559
560  batch_normalization_59 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_58[0][0]']
561  ormalization
    )

562

563  activation_59 (Activation)     (None, 16, 16, 96
    )   0           ['batch_normalization_59[0][0]']
564

565  conv2d_59 (Conv2D)             (None, 16, 16, 24
    )   20736       ['activation_59[0][0]']
566

567  concatenate_27 (Concatenate)   (None, 16, 16, 264
    )  0           ['concatenate_26[0][0]',
568
                    'conv2d_59[0][0]']
569

570  batch_normalization_60 (BatchN  (None, 16, 16, 264
    )  1056        ['concatenate_27[0][0]']
571  ormalization
    )

572

573  activation_60 (Activation)     (None, 16, 16, 264
    )  0           ['batch_normalization_60[0][0]']
574

575  conv2d_60 (Conv2D)             (None, 16, 16, 96
    )   25344       ['activation_60[0][0]']
576

577  batch_normalization_61 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_60[0][0]']
578  ormalization
    )
```

```
578
579

580  activation_61 (Activation)     (None, 16, 16, 96
    )   0           ['batch_normalization_61[0][0]']
581

582  conv2d_61 (Conv2D)             (None, 16, 16, 24
    )   20736       ['activation_61[0][0]']
583

584  concatenate_28 (Concatenate)   (None, 16, 16, 288
    )  0           ['concatenate_27[0][0]',
585
                   'conv2d_61[0][0]']
586

587  batch_normalization_62 (BatchN  (None, 16, 16, 288
    )  1152       ['concatenate_28[0][0]']
588  ormalization
    )

589

590  activation_62 (Activation)     (None, 16, 16, 288
    )  0           ['batch_normalization_62[0][0]']
591

592  conv2d_62 (Conv2D)             (None, 16, 16, 96
    )   27648       ['activation_62[0][0]']
593

594  batch_normalization_63 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_62[0][0]']
595  ormalization
    )

596

597  activation_63 (Activation)     (None, 16, 16, 96
    )   0           ['batch_normalization_63[0][0]']
```

```
598
599  conv2d_63 (Conv2D)              (None, 16, 16, 24
    )   20736       ['activation_63[0][0]']
600
601  concatenate_29 (Concatenate)   (None, 16, 16, 312
    )  0           ['concatenate_28[0][0]',
602
                    'conv2d_63[0][0]']
603
604  batch_normalization_64 (BatchN  (None, 16, 16, 312
    )  1248       ['concatenate_29[0][0]']
605  ormalization
    )
606
607  activation_64 (Activation)     (None, 16, 16, 312
    )  0           ['batch_normalization_64[0][0]']
608
609  conv2d_64 (Conv2D)              (None, 16, 16, 96
    )   29952       ['activation_64[0][0]']
610
611  batch_normalization_65 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_64[0][0]']
612  ormalization
    )
613
614  activation_65 (Activation)     (None, 16, 16, 96
    )   0           ['batch_normalization_65[0][0]']
615
616  conv2d_65 (Conv2D)              (None, 16, 16, 24
    )   20736       ['activation_65[0][0]']
617
```

```
617
618  concatenate_30 (Concatenate)   (None, 16, 16, 336
    )  0           ['concatenate_29[0][0]',
619
                    'conv2d_65[0][0]']
620

621  batch_normalization_66 (BatchN  (None, 16, 16, 336
    )  1344        ['concatenate_30[0][0]']
622  ormalization
    )

623

624  activation_66 (Activation)     (None, 16, 16, 336
    )  0           ['batch_normalization_66[0][0]']
625

626  conv2d_66 (Conv2D)             (None, 16, 16, 96
    )   32256       ['activation_66[0][0]']
627

628  batch_normalization_67 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_66[0][0]']
629  ormalization
    )

630

631  activation_67 (Activation)     (None, 16, 16, 96
    )   0           ['batch_normalization_67[0][0]']
632

633  conv2d_67 (Conv2D)             (None, 16, 16, 24
    )   20736       ['activation_67[0][0]']
634

635  concatenate_31 (Concatenate)   (None, 16, 16, 360
    )  0           ['concatenate_30[0][0]',
636
                    'conv2d_67[0][0]']
```

```
637
638  batch_normalization_68 (BatchN  (None, 16, 16, 360
    )  1440       ['concatenate_31[0][0]']
639  ormalization
    )
640
641  activation_68 (Activation)     (None, 16, 16, 360
    )  0           ['batch_normalization_68[0][0]']
642
643  conv2d_68 (Conv2D)             (None, 16, 16, 96
    )   34560       ['activation_68[0][0]']
644
645  batch_normalization_69 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_68[0][0]']
646  ormalization
    )
647
648  activation_69 (Activation)     (None, 16, 16, 96
    )   0           ['batch_normalization_69[0][0]']
649
650  conv2d_69 (Conv2D)             (None, 16, 16, 24
    )   20736       ['activation_69[0][0]']
651
652  concatenate_32 (Concatenate)   (None, 16, 16, 384
    )  0           ['concatenate_31[0][0]',
653
                    'conv2d_69[0][0]']
654
655  batch_normalization_70 (BatchN  (None, 16, 16, 384
    )  1536       ['concatenate_32[0][0]']
656  ormalization
```

```
656
    )

657

658  activation_70 (Activation)     (None, 16, 16, 384
    )  0           ['batch_normalization_70[0][0]']
659

660  conv2d_70 (Conv2D)             (None, 16, 16, 96
    )   36864       ['activation_70[0][0]']
661

662  batch_normalization_71 (BatchN  (None, 16, 16, 96
    )  384         ['conv2d_70[0][0]']
663  ormalization
    )

664

665  activation_71 (Activation)     (None, 16, 16, 96
    )   0           ['batch_normalization_71[0][0]']
666

667  conv2d_71 (Conv2D)             (None, 16, 16, 24
    )   20736       ['activation_71[0][0]']
668

669  concatenate_33 (Concatenate)   (None, 16, 16, 408
    )  0           ['concatenate_32[0][0]',
670
                    'conv2d_71[0][0]']
671

672  batch_normalization_72 (BatchN  (None, 16, 16, 408
    )  1632        ['concatenate_33[0][0]']
673  ormalization
    )

674
```

```
activation_72 (Activation)     (None, 16, 16, 408
)  0           ['batch_normalization_72[0][0]']


global_average_pooling2d (Glob  (None, 408
)         0           ['activation_72[0][0
]']
alAveragePooling2D
)


dropout (Dropout)              (None, 408
)          0           ['global_average_pooling2d[0
][0]'

               ]


dense (Dense)                  (None, 3
)            1227        ['dropout[0][0
]']


================================================
================================================
Total params: 1,659,291
Trainable params: 1,634,427
Non-trainable params: 24,864
________________________________________________
________________________________________________
Epoch 1/500
2026-05-22 19:10:09.314829: I tensorflow/
stream_executor/cuda/cuda_dnn.cc:384] Loaded cuDNN
version 8100
693/693 [==============================] - 216s
297ms/step - loss: 0.9040 - accuracy: 0.5716 -
val_loss: 1.7095 - val_accuracy: 0.4286 - lr: 0.0100
Epoch 2/500
693/693 [==============================] - 201s
288ms/step - loss: 0.7898 - accuracy: 0.6356 -
```

```
694 val_loss: 1.1803 - val_accuracy: 0.6140 - lr: 0.0100
695 Epoch 3/500
696 693/693 [==============================] - 200s
    288ms/step - loss: 0.7247 - accuracy: 0.6658 -
    val_loss: 1.0351 - val_accuracy: 0.5598 - lr: 0.0100
697 Epoch 4/500
698 693/693 [==============================] - 202s
    291ms/step - loss: 0.6588 - accuracy: 0.7007 -
    val_loss: 1.4993 - val_accuracy: 0.5504 - lr: 0.0100
699 Epoch 5/500
700 693/693 [==============================] - 204s
    293ms/step - loss: 0.6232 - accuracy: 0.7239 -
    val_loss: 1.5136 - val_accuracy: 0.4290 - lr: 0.0100
701 Epoch 6/500
702 693/693 [==============================] - 201s
    289ms/step - loss: 0.5752 - accuracy: 0.7460 -
    val_loss: 0.8334 - val_accuracy: 0.6509 - lr: 0.0100
703 Epoch 7/500
704 693/693 [==============================] - 200s
    287ms/step - loss: 0.5354 - accuracy: 0.7660 -
    val_loss: 0.6753 - val_accuracy: 0.6784 - lr: 0.0100
705 Epoch 8/500
706 693/693 [==============================] - 200s
    288ms/step - loss: 0.5039 - accuracy: 0.7834 -
    val_loss: 2.0002 - val_accuracy: 0.5410 - lr: 0.0100
707 Epoch 9/500
708 693/693 [==============================] - 199s
    286ms/step - loss: 0.4744 - accuracy: 0.7971 -
    val_loss: 0.4990 - val_accuracy: 0.7904 - lr: 0.0100
709 Epoch 10/500
710 693/693 [==============================] - 199s
    286ms/step - loss: 0.4580 - accuracy: 0.8081 -
    val_loss: 0.6633 - val_accuracy: 0.7199 - lr: 0.0100
711 Epoch 11/500
712 693/693 [==============================] - 199s
    286ms/step - loss: 0.4294 - accuracy: 0.8216 -
    val_loss: 1.3382 - val_accuracy: 0.5544 - lr: 0.0100
713 Epoch 12/500
714 693/693 [==============================] - 199s
    286ms/step - loss: 0.4156 - accuracy: 0.8248 -
    val_loss: 0.4128 - val_accuracy: 0.8291 - lr: 0.0100
```

```
715 Epoch 13/500
716 693/693 [==============================] - 198s
    285ms/step - loss: 0.3963 - accuracy: 0.8372 -
    val_loss: 0.6984 - val_accuracy: 0.7105 - lr: 0.0100
717 Epoch 14/500
718 693/693 [==============================] - 198s
    285ms/step - loss: 0.3822 - accuracy: 0.8438 -
    val_loss: 1.2967 - val_accuracy: 0.6249 - lr: 0.0100
719 Epoch 15/500
720 693/693 [==============================] - 198s
    285ms/step - loss: 0.3670 - accuracy: 0.8537 -
    val_loss: 0.6521 - val_accuracy: 0.7651 - lr: 0.0100
721 Epoch 16/500
722 693/693 [==============================] - 199s
    286ms/step - loss: 0.3679 - accuracy: 0.8474 -
    val_loss: 0.6284 - val_accuracy: 0.7763 - lr: 0.0100
723 Epoch 17/500
724 693/693 [==============================] - 200s
    287ms/step - loss: 0.3426 - accuracy: 0.8580 -
    val_loss: 0.3890 - val_accuracy: 0.8478 - lr: 0.0100
725 Epoch 18/500
726 693/693 [==============================] - 200s
    287ms/step - loss: 0.3402 - accuracy: 0.8633 -
    val_loss: 1.8977 - val_accuracy: 0.5641 - lr: 0.0100
727 Epoch 19/500
728 693/693 [==============================] - 199s
    286ms/step - loss: 0.3202 - accuracy: 0.8699 -
    val_loss: 0.4675 - val_accuracy: 0.8197 - lr: 0.0100
729 Epoch 20/500
730 693/693 [==============================] - 200s
    288ms/step - loss: 0.3086 - accuracy: 0.8787 -
    val_loss: 0.7908 - val_accuracy: 0.6834 - lr: 0.0100
731 Epoch 21/500
732 693/693 [==============================] - 200s
    287ms/step - loss: 0.3066 - accuracy: 0.8771 -
    val_loss: 0.5807 - val_accuracy: 0.7832 - lr: 0.0100
733 Epoch 22/500
734 693/693 [==============================] - 200s
    286ms/step - loss: 0.2901 - accuracy: 0.8827 -
    val_loss: 0.4255 - val_accuracy: 0.8171 - lr: 0.0100
735 Epoch 23/500
```

```
736 693/693 [==============================] - 199s
    287ms/step - loss: 0.2842 - accuracy: 0.8846 -
    val_loss: 0.5250 - val_accuracy: 0.8124 - lr: 0.0100
737 Epoch 24/500
738 693/693 [==============================] - 199s
    286ms/step - loss: 0.2763 - accuracy: 0.8921 -
    val_loss: 0.5232 - val_accuracy: 0.7940 - lr: 0.0100
739 Epoch 25/500
740 693/693 [==============================] - 198s
    286ms/step - loss: 0.2577 - accuracy: 0.8978 -
    val_loss: 0.4154 - val_accuracy: 0.8417 - lr: 0.0100
741 Epoch 26/500
742 693/693 [==============================] - 199s
    286ms/step - loss: 0.2573 - accuracy: 0.8968 -
    val_loss: 0.7068 - val_accuracy: 0.7474 - lr: 0.0100
743 Epoch 27/500
744 693/693 [==============================] - 199s
    286ms/step - loss: 0.2591 - accuracy: 0.8976 -
    val_loss: 0.6979 - val_accuracy: 0.7477 - lr: 0.0100
745 Epoch 28/500
746 693/693 [==============================] - 199s
    286ms/step - loss: 0.2439 - accuracy: 0.9044 -
    val_loss: 0.4374 - val_accuracy: 0.8244 - lr: 0.0100
747 Epoch 29/500
748 693/693 [==============================] - 199s
    286ms/step - loss: 0.2554 - accuracy: 0.8990 -
    val_loss: 0.4179 - val_accuracy: 0.8287 - lr: 0.0100
749 Epoch 30/500
750 692/693 [============================>.] - ETA: 0s
     - loss: 0.2369 - accuracy: 0.9085
751 [Plateau Check] Variação últimas 30 épocas: 0.419227
752 693/693 [==============================] - 199s
    286ms/step - loss: 0.2371 - accuracy: 0.9084 -
    val_loss: 0.5808 - val_accuracy: 0.7897 - lr: 0.0100
753 Epoch 31/500
754 692/693 [============================>.] - ETA: 0s
     - loss: 0.2211 - accuracy: 0.9109
755 [Plateau Check] Variação últimas 30 épocas: 0.458619
756 693/693 [==============================] - 199s
    286ms/step - loss: 0.2214 - accuracy: 0.9108 -
    val_loss: 0.3115 - val_accuracy: 0.8876 - lr: 0.0100
```

```
Epoch 32/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.2236 - accuracy: 0.9122
[Plateau Check] Variação últimas 30 épocas: 0.458619
693/693 [==============================] - 199s
286ms/step - loss: 0.2236 - accuracy: 0.9122 -
val_loss: 0.4848 - val_accuracy: 0.7991 - lr: 0.0100
Epoch 33/500
693/693 [==============================] - ETA: 0s
 - loss: 0.2185 - accuracy: 0.9151
[Plateau Check] Variação últimas 30 épocas: 0.458619
693/693 [==============================] - 199s
286ms/step - loss: 0.2185 - accuracy: 0.9151 -
val_loss: 0.3892 - val_accuracy: 0.8518 - lr: 0.0100
Epoch 34/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.2055 - accuracy: 0.9222
[Plateau Check] Variação últimas 30 épocas: 0.458619
693/693 [==============================] - 199s
286ms/step - loss: 0.2056 - accuracy: 0.9222 -
val_loss: 0.5394 - val_accuracy: 0.8048 - lr: 0.0100
Epoch 35/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.2003 - accuracy: 0.9269
[Plateau Check] Variação últimas 30 épocas: 0.347669
693/693 [==============================] - 199s
286ms/step - loss: 0.2005 - accuracy: 0.9268 -
val_loss: 0.3192 - val_accuracy: 0.8887 - lr: 0.0100
Epoch 36/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.2008 - accuracy: 0.9200
[Plateau Check] Variação últimas 30 épocas: 0.347669
693/693 [==============================] - 199s
286ms/step - loss: 0.2009 - accuracy: 0.9199 -
val_loss: 0.3700 - val_accuracy: 0.8670 - lr: 0.0100
Epoch 37/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.1992 - accuracy: 0.9246
[Plateau Check] Variação últimas 30 épocas: 0.347669
693/693 [==============================] - 198s
286ms/step - loss: 0.1991 - accuracy: 0.9246 -
```

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780 val_loss: 0.5443 - val_accuracy: 0.7868 - lr: 0.0100
781 Epoch 38/500
782 693/693 [==============================] - ETA: 0s
     - loss: 0.1903 - accuracy: 0.9249
783 [Plateau Check] Variação últimas 30 épocas: 0.334297
784 693/693 [==============================] - 199s
    286ms/step - loss: 0.1903 - accuracy: 0.9249 -
    val_loss: 0.5681 - val_accuracy: 0.7795 - lr: 0.0100
785 Epoch 39/500
786 692/693 [==============================>.] - ETA: 0s
     - loss: 0.1924 - accuracy: 0.9243
787 [Plateau Check] Variação últimas 30 épocas: 0.342248
788 693/693 [==============================] - 199s
    286ms/step - loss: 0.1924 - accuracy: 0.9243 -
    val_loss: 0.3248 - val_accuracy: 0.8966 - lr: 0.0100
789 Epoch 40/500
790 692/693 [==============================>.] - ETA: 0s
     - loss: 0.1793 - accuracy: 0.9296
791 [Plateau Check] Variação últimas 30 épocas: 0.342248
792 693/693 [==============================] - 199s
    286ms/step - loss: 0.1793 - accuracy: 0.9297 -
    val_loss: 0.6161 - val_accuracy: 0.7911 - lr: 0.0100
793 Epoch 41/500
794 692/693 [==============================>.] - ETA: 0s
     - loss: 0.1768 - accuracy: 0.9314
795 [Plateau Check] Variação últimas 30 épocas: 0.332490
796 693/693 [==============================] - 198s
    285ms/step - loss: 0.1770 - accuracy: 0.9314 -
    val_loss: 0.5027 - val_accuracy: 0.8334 - lr: 0.0100
797 Epoch 42/500
798 692/693 [==============================>.] - ETA: 0s
     - loss: 0.1695 - accuracy: 0.9348
799 [Plateau Check] Variação últimas 30 épocas: 0.332490
800 693/693 [==============================] - 198s
    285ms/step - loss: 0.1694 - accuracy: 0.9348 -
    val_loss: 0.4471 - val_accuracy: 0.8544 - lr: 0.0100
801 Epoch 43/500
802 692/693 [==============================>.] - ETA: 0s
     - loss: 0.1549 - accuracy: 0.9401
803 [Plateau Check] Variação últimas 30 épocas: 0.332490
804 693/693 [==============================] - 198s
```

```
804 285ms/step - loss: 0.1549 - accuracy: 0.9401 -
    val_loss: 0.3789 - val_accuracy: 0.8630 - lr: 0.0100
805 Epoch 44/500
806 692/693 [============================>.] - ETA: 0s
     - loss: 0.1534 - accuracy: 0.9421
807 [Plateau Check] Variação últimas 30 épocas: 0.332490
808 693/693 [==============================] - 198s
    285ms/step - loss: 0.1533 - accuracy: 0.9421 -
    val_loss: 0.4005 - val_accuracy: 0.8699 - lr: 0.0100
809 Epoch 45/500
810 692/693 [============================>.] - ETA: 0s
     - loss: 0.1498 - accuracy: 0.9438
811 [Plateau Check] Variação últimas 30 épocas: 0.332490
812 693/693 [==============================] - 198s
    286ms/step - loss: 0.1498 - accuracy: 0.9437 -
    val_loss: 0.6522 - val_accuracy: 0.7933 - lr: 0.0100
813 Epoch 46/500
814 692/693 [============================>.] - ETA: 0s
     - loss: 0.1517 - accuracy: 0.9426
815 [Plateau Check] Variação últimas 30 épocas: 0.332490
816 693/693 [==============================] - 199s
    286ms/step - loss: 0.1517 - accuracy: 0.9426 -
    val_loss: 0.4391 - val_accuracy: 0.8435 - lr: 0.0100
817 Epoch 47/500
818 692/693 [============================>.] - ETA: 0s
     - loss: 0.0919 - accuracy: 0.9687
819 [Plateau Check] Variação últimas 30 épocas: 0.349837
820 693/693 [==============================] - 199s
    286ms/step - loss: 0.0919 - accuracy: 0.9687 -
    val_loss: 0.2640 - val_accuracy: 0.9140 - lr: 1.
    0000e-03
821 Epoch 48/500
822 692/693 [============================>.] - ETA: 0s
     - loss: 0.0713 - accuracy: 0.9760
823 [Plateau Check] Variação últimas 30 épocas: 0.232743
824 693/693 [==============================] - 198s
    285ms/step - loss: 0.0721 - accuracy: 0.9759 -
    val_loss: 0.2660 - val_accuracy: 0.9162 - lr: 1.
    0000e-03
825 Epoch 49/500
826 692/693 [============================>.] - ETA: 0s
```

```
826  - loss: 0.0637 - accuracy: 0.9780
827 [Plateau Check] Variação últimas 30 épocas: 0.236718
828 693/693 [==============================] - 198s
    285ms/step - loss: 0.0638 - accuracy: 0.9780 -
    val_loss: 0.2725 - val_accuracy: 0.9201 - lr: 1.
    0000e-03
829 Epoch 50/500
830 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0572 - accuracy: 0.9810
831 [Plateau Check] Variação últimas 30 épocas: 0.172750
832 693/693 [==============================] - 198s
    285ms/step - loss: 0.0573 - accuracy: 0.9809 -
    val_loss: 0.2773 - val_accuracy: 0.9165 - lr: 1.
    0000e-03
833 Epoch 51/500
834 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0527 - accuracy: 0.9822
835 [Plateau Check] Variação últimas 30 épocas: 0.172750
836 693/693 [==============================] - 198s
    285ms/step - loss: 0.0526 - accuracy: 0.9822 -
    val_loss: 0.2842 - val_accuracy: 0.9180 - lr: 1.
    0000e-03
837 Epoch 52/500
838 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0518 - accuracy: 0.9824
839 [Plateau Check] Variação últimas 30 épocas: 0.172750
840 693/693 [==============================] - 198s
    285ms/step - loss: 0.0518 - accuracy: 0.9824 -
    val_loss: 0.2883 - val_accuracy: 0.9187 - lr: 1.
    0000e-03
841 Epoch 53/500
842 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0524 - accuracy: 0.9827
843 [Plateau Check] Variação últimas 30 épocas: 0.172750
844 693/693 [==============================] - 199s
    286ms/step - loss: 0.0526 - accuracy: 0.9827 -
    val_loss: 0.2853 - val_accuracy: 0.9172 - lr: 1.
    0000e-03
845 Epoch 54/500
846 693/693 [==============================] - ETA: 0s
     - loss: 0.0449 - accuracy: 0.9857
```

```
847 [Plateau Check] Variação últimas 30 épocas: 0.172750
848 693/693 [==============================] - 198s
    285ms/step - loss: 0.0449 - accuracy: 0.9857 -
    val_loss: 0.2863 - val_accuracy: 0.9176 - lr: 1.
    0000e-03
849 Epoch 55/500
850 693/693 [==============================] - ETA: 0s
     - loss: 0.0437 - accuracy: 0.9847
851 [Plateau Check] Variação últimas 30 épocas: 0.172750
852 693/693 [==============================] - 198s
    285ms/step - loss: 0.0437 - accuracy: 0.9847 -
    val_loss: 0.2831 - val_accuracy: 0.9190 - lr: 1.
    0000e-03
853 Epoch 56/500
854 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0415 - accuracy: 0.9877
855 [Plateau Check] Variação últimas 30 épocas: 0.176003
856 693/693 [==============================] - 198s
    285ms/step - loss: 0.0415 - accuracy: 0.9877 -
    val_loss: 0.2962 - val_accuracy: 0.9237 - lr: 1.
    0000e-03
857 Epoch 57/500
858 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0417 - accuracy: 0.9872
859 [Plateau Check] Variação últimas 30 épocas: 0.144199
860 693/693 [==============================] - 198s
    285ms/step - loss: 0.0417 - accuracy: 0.9872 -
    val_loss: 0.2938 - val_accuracy: 0.9227 - lr: 1.
    0000e-03
861 Epoch 58/500
862 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0375 - accuracy: 0.9876
863 [Plateau Check] Variação últimas 30 épocas: 0.144199
864 693/693 [==============================] - 199s
    286ms/step - loss: 0.0375 - accuracy: 0.9876 -
    val_loss: 0.2986 - val_accuracy: 0.9212 - lr: 1.
    0000e-03
865 Epoch 59/500
866 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0362 - accuracy: 0.9885
867 [Plateau Check] Variação últimas 30 épocas: 0.144199
```

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868 693/693 [==============================] - 199s
    286ms/step - loss: 0.0362 - accuracy: 0.9885 -
    val_loss: 0.3011 - val_accuracy: 0.9201 - lr: 1.
    0000e-03
869 Epoch 60/500
870 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0404 - accuracy: 0.9877
871 [Plateau Check] Variação últimas 30 épocas: 0.144199
872 693/693 [==============================] - 199s
    286ms/step - loss: 0.0415 - accuracy: 0.9875 -
    val_loss: 0.3048 - val_accuracy: 0.9194 - lr: 1.
    0000e-03
873 Epoch 61/500
874 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0370 - accuracy: 0.9883
875 [Plateau Check] Variação últimas 30 épocas: 0.144199
876 693/693 [==============================] - 198s
    285ms/step - loss: 0.0372 - accuracy: 0.9883 -
    val_loss: 0.3049 - val_accuracy: 0.9205 - lr: 1.
    0000e-03
877 Epoch 62/500
878 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0378 - accuracy: 0.9888
879 [Plateau Check] Variação últimas 30 épocas: 0.144199
880 693/693 [==============================] - 198s
    285ms/step - loss: 0.0378 - accuracy: 0.9888 -
    val_loss: 0.3074 - val_accuracy: 0.9165 - lr: 1.
    0000e-03
881 Epoch 63/500
882 693/693 [==============================] - ETA: 0s
     - loss: 0.0337 - accuracy: 0.9896
883 [Plateau Check] Variação últimas 30 épocas: 0.144199
884 693/693 [==============================] - 198s
    285ms/step - loss: 0.0337 - accuracy: 0.9896 -
    val_loss: 0.3045 - val_accuracy: 0.9183 - lr: 1.
    0000e-04
885 Epoch 64/500
886 693/693 [==============================] - ETA: 0s
     - loss: 0.0341 - accuracy: 0.9895
887 [Plateau Check] Variação últimas 30 épocas: 0.144199
888 693/693 [==============================] - 198s
```

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888 285ms/step - loss: 0.0341 - accuracy: 0.9895 -
    val_loss: 0.3017 - val_accuracy: 0.9176 - lr: 1.
    0000e-04
889 Epoch 65/500
890 692/693 [============================>.] - ETA: 0s
     - loss: 0.0331 - accuracy: 0.9903
891 [Plateau Check] Variação últimas 30 épocas: 0.144199
892 693/693 [==============================] - 199s
    286ms/step - loss: 0.0332 - accuracy: 0.9903 -
    val_loss: 0.3009 - val_accuracy: 0.9187 - lr: 1.
    0000e-04
893 Epoch 66/500
894 692/693 [============================>.] - ETA: 0s
     - loss: 0.0309 - accuracy: 0.9911
895 [Plateau Check] Variação últimas 30 épocas: 0.144199
896 693/693 [==============================] - 198s
    285ms/step - loss: 0.0309 - accuracy: 0.9911 -
    val_loss: 0.3020 - val_accuracy: 0.9187 - lr: 1.
    0000e-04
897 Epoch 67/500
898 693/693 [==============================] - ETA: 0s
     - loss: 0.0306 - accuracy: 0.9912
899 [Plateau Check] Variação últimas 30 épocas: 0.144199
900 693/693 [==============================] - 199s
    285ms/step - loss: 0.0306 - accuracy: 0.9912 -
    val_loss: 0.3006 - val_accuracy: 0.9194 - lr: 1.
    0000e-04
901 Epoch 68/500
902 692/693 [============================>.] - ETA: 0s
     - loss: 0.0308 - accuracy: 0.9910
903 [Plateau Check] Variação últimas 30 épocas: 0.132635
904 693/693 [==============================] - 198s
    285ms/step - loss: 0.0308 - accuracy: 0.9910 -
    val_loss: 0.2995 - val_accuracy: 0.9209 - lr: 1.
    0000e-04
905 Epoch 69/500
906 693/693 [==============================] - ETA: 0s
     - loss: 0.0272 - accuracy: 0.9917
907 [Plateau Check] Variação últimas 30 épocas: 0.132635
908 693/693 [==============================] - 198s
    285ms/step - loss: 0.0272 - accuracy: 0.9917 -
```

```
908 val_loss: 0.2992 - val_accuracy: 0.9190 - lr: 1.
    0000e-04
909 Epoch 70/500
910 692/693 [============================>.] - ETA: 0s
     - loss: 0.0313 - accuracy: 0.9906
911 [Plateau Check] Variação últimas 30 épocas: 0.130466
912 693/693 [==============================] - 198s
    285ms/step - loss: 0.0323 - accuracy: 0.9904 -
    val_loss: 0.3016 - val_accuracy: 0.9209 - lr: 1.
    0000e-04
913 Epoch 71/500
914 692/693 [============================>.] - ETA: 0s
     - loss: 0.0305 - accuracy: 0.9909
915 [Plateau Check] Variação últimas 30 épocas: 0.130466
916 693/693 [==============================] - 198s
    285ms/step - loss: 0.0309 - accuracy: 0.9908 -
    val_loss: 0.2990 - val_accuracy: 0.9176 - lr: 1.
    0000e-04
917 Epoch 72/500
918 692/693 [============================>.] - ETA: 0s
     - loss: 0.0306 - accuracy: 0.9911
919 [Plateau Check] Variação últimas 30 épocas: 0.130466
920 693/693 [==============================] - 199s
    286ms/step - loss: 0.0306 - accuracy: 0.9911 -
    val_loss: 0.2996 - val_accuracy: 0.9212 - lr: 1.
    0000e-04
921 Epoch 73/500
922 693/693 [==============================] - ETA: 0s
     - loss: 0.0286 - accuracy: 0.9926
923 [Plateau Check] Variação últimas 30 épocas: 0.130466
924 693/693 [==============================] - 199s
    286ms/step - loss: 0.0286 - accuracy: 0.9926 -
    val_loss: 0.3003 - val_accuracy: 0.9209 - lr: 1.
    0000e-04
925 Epoch 74/500
926 692/693 [============================>.] - ETA: 0s
     - loss: 0.0251 - accuracy: 0.9926
927 [Plateau Check] Variação últimas 30 épocas: 0.130466
928 693/693 [==============================] - 198s
    285ms/step - loss: 0.0251 - accuracy: 0.9926 -
    val_loss: 0.3017 - val_accuracy: 0.9194 - lr: 1.
```

```
928 0000e-04
929 Epoch 75/500
930 692/693 [============================>.] - ETA: 0s
     - loss: 0.0298 - accuracy: 0.9912
931 [Plateau Check] Variação últimas 30 épocas: 0.080231
932 693/693 [==============================] - 198s
    285ms/step - loss: 0.0298 - accuracy: 0.9912 -
    val_loss: 0.3017 - val_accuracy: 0.9205 - lr: 1.
    0000e-04
933 Epoch 76/500
934 692/693 [============================>.] - ETA: 0s
     - loss: 0.0315 - accuracy: 0.9907
935 [Plateau Check] Variação últimas 30 épocas: 0.009758
936 693/693 [==============================] - 198s
    285ms/step - loss: 0.0315 - accuracy: 0.9907 -
    val_loss: 0.3009 - val_accuracy: 0.9198 - lr: 1.
    0000e-04
937 Epoch 77/500
938 692/693 [============================>.] - ETA: 0s
     - loss: 0.0283 - accuracy: 0.9921
939 [Plateau Check] Variação últimas 30 épocas: 0.007589
940 693/693 [==============================] - 198s
    285ms/step - loss: 0.0283 - accuracy: 0.9921 -
    val_loss: 0.3013 - val_accuracy: 0.9209 - lr: 1.
    0000e-04
941 Epoch 78/500
942 692/693 [============================>.] - ETA: 0s
     - loss: 0.0277 - accuracy: 0.9919
943 [Plateau Check] Variação últimas 30 épocas: 0.007228
944 693/693 [==============================] - 198s
    285ms/step - loss: 0.0277 - accuracy: 0.9919 -
    val_loss: 0.3013 - val_accuracy: 0.9223 - lr: 1.
    0000e-05
945 Epoch 79/500
946 692/693 [============================>.] - ETA: 0s
     - loss: 0.0272 - accuracy: 0.9930
947 [Plateau Check] Variação últimas 30 épocas: 0.007228
948 693/693 [==============================] - 198s
    286ms/step - loss: 0.0272 - accuracy: 0.9930 -
    val_loss: 0.3004 - val_accuracy: 0.9212 - lr: 1.
    0000e-05
```

```
949 Epoch 80/500
950 693/693 [==============================] - ETA: 0s
     - loss: 0.0243 - accuracy: 0.9930
951 [Plateau Check] Variação últimas 30 épocas: 0.007228
952 693/693 [==============================] - 198s
    285ms/step - loss: 0.0243 - accuracy: 0.9930 -
    val_loss: 0.2986 - val_accuracy: 0.9209 - lr: 1.
    0000e-05
953 Epoch 81/500
954 693/693 [==============================] - ETA: 0s
     - loss: 0.0283 - accuracy: 0.9912
955 [Plateau Check] Variação últimas 30 épocas: 0.007228
956 693/693 [==============================] - 198s
    285ms/step - loss: 0.0283 - accuracy: 0.9912 -
    val_loss: 0.3009 - val_accuracy: 0.9216 - lr: 1.
    0000e-05
957 Epoch 82/500
958 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0267 - accuracy: 0.9929
959 [Plateau Check] Variação últimas 30 épocas: 0.007228
960 693/693 [==============================] - 198s
    285ms/step - loss: 0.0271 - accuracy: 0.9928 -
    val_loss: 0.3003 - val_accuracy: 0.9223 - lr: 1.
    0000e-05
961 Epoch 83/500
962 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0277 - accuracy: 0.9929
963 [Plateau Check] Variação últimas 30 épocas: 0.007228
964 693/693 [==============================] - 199s
    286ms/step - loss: 0.0280 - accuracy: 0.9928 -
    val_loss: 0.3023 - val_accuracy: 0.9212 - lr: 1.
    0000e-05
965 Epoch 84/500
966 692/693 [==============================>.] - ETA: 0s
     - loss: 0.0252 - accuracy: 0.9929
967 [Plateau Check] Variação últimas 30 épocas: 0.007228
968 693/693 [==============================] - 198s
    285ms/step - loss: 0.0256 - accuracy: 0.9928 -
    val_loss: 0.3013 - val_accuracy: 0.9205 - lr: 1.
    0000e-05
969 Epoch 85/500
```

```
970 692/693 [============================>.] - ETA: 0s
     - loss: 0.0286 - accuracy: 0.9910
971 [Plateau Check] Variação últimas 30 épocas: 0.
    007228
972 693/693 [==============================] - 198s
    285ms/step - loss: 0.0288 - accuracy: 0.9909 -
    val_loss: 0.3007 - val_accuracy: 0.9212 - lr: 1.
    0000e-05
973 Epoch 86/500
974 692/693 [============================>.] - ETA: 0s
     - loss: 0.0280 - accuracy: 0.9919
975 [Plateau Check] Variação últimas 30 épocas: 0.
    006144
976 693/693 [==============================] - 198s
    285ms/step - loss: 0.0282 - accuracy: 0.9918 -
    val_loss: 0.3014 - val_accuracy: 0.9223 - lr: 1.
    0000e-05
977 Epoch 87/500
978 692/693 [============================>.] - ETA: 0s
     - loss: 0.0297 - accuracy: 0.9912
979 [Plateau Check] Variação últimas 30 épocas: 0.
    005782
980 693/693 [==============================] - 198s
    285ms/step - loss: 0.0301 - accuracy: 0.9912 -
    val_loss: 0.3036 - val_accuracy: 0.9209 - lr: 1.
    0000e-05
981 Epoch 88/500
982 692/693 [============================>.] - ETA: 0s
     - loss: 0.0284 - accuracy: 0.9912
983 [Plateau Check] Variação últimas 30 épocas: 0.
    005782
984 693/693 [==============================] - 198s
    285ms/step - loss: 0.0286 - accuracy: 0.9912 -
    val_loss: 0.3020 - val_accuracy: 0.9201 - lr: 1.
    0000e-05
985 Epoch 89/500
986 692/693 [============================>.] - ETA: 0s
     - loss: 0.0238 - accuracy: 0.9936
987 [Plateau Check] Variação últimas 30 épocas: 0.
    005782
988 693/693 [==============================] - 198s
```

```
 988 285ms/step - loss: 0.0240 - accuracy: 0.9935 -
     val_loss: 0.3011 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
 989 Epoch 90/500
 990 692/693 [============================>.] - ETA: 0s
      - loss: 0.0284 - accuracy: 0.9915
 991 [Plateau Check] Variação últimas 30 épocas: 0.
     005782
 992 693/693 [==============================] - 198s
     285ms/step - loss: 0.0283 - accuracy: 0.9915 -
     val_loss: 0.3026 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
 993 Epoch 91/500
 994 692/693 [============================>.] - ETA: 0s
      - loss: 0.0279 - accuracy: 0.9925
 995 [Plateau Check] Variação últimas 30 épocas: 0.
     005782
 996 693/693 [==============================] - 198s
     285ms/step - loss: 0.0279 - accuracy: 0.9924 -
     val_loss: 0.3021 - val_accuracy: 0.9216 - lr: 1.
     0000e-05
 997 Epoch 92/500
 998 692/693 [============================>.] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9923
 999 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1000 693/693 [==============================] - 198s
     285ms/step - loss: 0.0268 - accuracy: 0.9923 -
     val_loss: 0.3021 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1001 Epoch 93/500
1002 693/693 [==============================] - ETA: 0s
      - loss: 0.0278 - accuracy: 0.9915
1003 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1004 693/693 [==============================] - 198s
     285ms/step - loss: 0.0278 - accuracy: 0.9915 -
     val_loss: 0.3023 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1005 Epoch 94/500
1006 692/693 [============================>.] - ETA: 0s
```

```
1006  - loss: 0.0309 - accuracy: 0.9905
1007 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1008 693/693 [==============================] - 198s
     285ms/step - loss: 0.0309 - accuracy: 0.9905 -
     val_loss: 0.3005 - val_accuracy: 0.9219 - lr: 1.
     0000e-05
1009 Epoch 95/500
1010 692/693 [============================>.] - ETA: 0s
      - loss: 0.0289 - accuracy: 0.9916
1011 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1012 693/693 [==============================] - 198s
     285ms/step - loss: 0.0293 - accuracy: 0.9915 -
     val_loss: 0.3010 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1013 Epoch 96/500
1014 693/693 [==============================] - ETA: 0s
      - loss: 0.0279 - accuracy: 0.9925
1015 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1016 693/693 [==============================] - 198s
     285ms/step - loss: 0.0279 - accuracy: 0.9925 -
     val_loss: 0.3025 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1017 Epoch 97/500
1018 692/693 [============================>.] - ETA: 0s
      - loss: 0.0276 - accuracy: 0.9919
1019 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1020 693/693 [==============================] - 198s
     285ms/step - loss: 0.0278 - accuracy: 0.9918 -
     val_loss: 0.3029 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1021 Epoch 98/500
1022 692/693 [============================>.] - ETA: 0s
      - loss: 0.0243 - accuracy: 0.9937
1023 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1024 693/693 [==============================] - 198s
     285ms/step - loss: 0.0243 - accuracy: 0.9937 -
```

```
1024 val_loss: 0.3015 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1025 Epoch 99/500
1026 692/693 [============================>.] - ETA: 0s
      - loss: 0.0278 - accuracy: 0.9923
1027 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1028 693/693 [==============================] - 198s
     285ms/step - loss: 0.0278 - accuracy: 0.9923 -
     val_loss: 0.3018 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1029 Epoch 100/500
1030 692/693 [============================>.] - ETA: 0s
      - loss: 0.0273 - accuracy: 0.9921
1031 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1032 693/693 [==============================] - 198s
     285ms/step - loss: 0.0273 - accuracy: 0.9921 -
     val_loss: 0.3024 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1033 Epoch 101/500
1034 692/693 [============================>.] - ETA: 0s
      - loss: 0.0260 - accuracy: 0.9925
1035 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1036 693/693 [==============================] - 198s
     285ms/step - loss: 0.0260 - accuracy: 0.9925 -
     val_loss: 0.3036 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1037 Epoch 102/500
1038 693/693 [==============================] - ETA: 0s
      - loss: 0.0294 - accuracy: 0.9916
1039 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1040 693/693 [==============================] - 198s
     285ms/step - loss: 0.0294 - accuracy: 0.9916 -
     val_loss: 0.3025 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1041 Epoch 103/500
1042 693/693 [==============================] - ETA: 0s
      - loss: 0.0256 - accuracy: 0.9921
```

```
1043 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1044 693/693 [==============================] - 199s
     285ms/step - loss: 0.0256 - accuracy: 0.9921 -
     val_loss: 0.3015 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1045 Epoch 104/500
1046 692/693 [============================>.] - ETA: 0s
      - loss: 0.0307 - accuracy: 0.9909
1047 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1048 693/693 [==============================] - 198s
     285ms/step - loss: 0.0307 - accuracy: 0.9909 -
     val_loss: 0.3021 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1049 Epoch 105/500
1050 692/693 [============================>.] - ETA: 0s
      - loss: 0.0278 - accuracy: 0.9922
1051 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1052 693/693 [==============================] - 198s
     285ms/step - loss: 0.0278 - accuracy: 0.9922 -
     val_loss: 0.3007 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1053 Epoch 106/500
1054 692/693 [============================>.] - ETA: 0s
      - loss: 0.0275 - accuracy: 0.9921
1055 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1056 693/693 [==============================] - 198s
     285ms/step - loss: 0.0275 - accuracy: 0.9921 -
     val_loss: 0.3008 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1057 Epoch 107/500
1058 692/693 [============================>.] - ETA: 0s
      - loss: 0.0282 - accuracy: 0.9918
1059 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1060 693/693 [==============================] - 198s
     285ms/step - loss: 0.0296 - accuracy: 0.9916 -
     val_loss: 0.3015 - val_accuracy: 0.9212 - lr: 1.
```

```
1060 0000e-05
1061 Epoch 108/500
1062 692/693 [============================>.] - ETA: 0s
      - loss: 0.0276 - accuracy: 0.9917
1063 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1064 693/693 [==============================] - 198s
     285ms/step - loss: 0.0276 - accuracy: 0.9917 -
     val_loss: 0.3015 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1065 Epoch 109/500
1066 692/693 [============================>.] - ETA: 0s
      - loss: 0.0282 - accuracy: 0.9922
1067 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1068 693/693 [==============================] - 198s
     285ms/step - loss: 0.0282 - accuracy: 0.9922 -
     val_loss: 0.3015 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1069 Epoch 110/500
1070 692/693 [============================>.] - ETA: 0s
      - loss: 0.0278 - accuracy: 0.9925
1071 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1072 693/693 [==============================] - 198s
     285ms/step - loss: 0.0280 - accuracy: 0.9924 -
     val_loss: 0.3012 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1073 Epoch 111/500
1074 692/693 [============================>.] - ETA: 0s
      - loss: 0.0257 - accuracy: 0.9934
1075 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1076 693/693 [==============================] - 198s
     285ms/step - loss: 0.0257 - accuracy: 0.9934 -
     val_loss: 0.3018 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1077 Epoch 112/500
1078 693/693 [==============================] - ETA: 0s
      - loss: 0.0278 - accuracy: 0.9921
1079 [Plateau Check] Variação últimas 30 épocas: 0.
```

```
004337
693/693 [==============================] - 198s
285ms/step - loss: 0.0278 - accuracy: 0.9921 -
val_loss: 0.3002 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 113/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0277 - accuracy: 0.9916
[Plateau Check] Variação últimas 30 épocas: 0.
004337
693/693 [==============================] - 198s
285ms/step - loss: 0.0277 - accuracy: 0.9916 -
val_loss: 0.3020 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 114/500
692/693 [==============================>.] - ETA: 0s
 - loss: 0.0258 - accuracy: 0.9930
[Plateau Check] Variação últimas 30 épocas: 0.
004337
693/693 [==============================] - 198s
285ms/step - loss: 0.0258 - accuracy: 0.9930 -
val_loss: 0.3018 - val_accuracy: 0.9201 - lr: 1.
0000e-05
Epoch 115/500
692/693 [==============================>.] - ETA: 0s
 - loss: 0.0260 - accuracy: 0.9929
[Plateau Check] Variação últimas 30 épocas: 0.
004337
693/693 [==============================] - 198s
285ms/step - loss: 0.0260 - accuracy: 0.9929 -
val_loss: 0.3022 - val_accuracy: 0.9201 - lr: 1.
0000e-05
Epoch 116/500
692/693 [==============================>.] - ETA: 0s
 - loss: 0.0254 - accuracy: 0.9927
[Plateau Check] Variação últimas 30 épocas: 0.
003975
693/693 [==============================] - 198s
285ms/step - loss: 0.0254 - accuracy: 0.9927 -
val_loss: 0.3015 - val_accuracy: 0.9194 - lr: 1.
0000e-05
```

```
Epoch 117/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0282 - accuracy: 0.9912
[Plateau Check] Variação últimas 30 épocas: 0.
004698
693/693 [==============================] - 199s
286ms/step - loss: 0.0282 - accuracy: 0.9912 -
val_loss: 0.3015 - val_accuracy: 0.9227 - lr: 1.
0000e-05
Epoch 118/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0286 - accuracy: 0.9925
[Plateau Check] Variação últimas 30 épocas: 0.
004698
693/693 [==============================] - 198s
285ms/step - loss: 0.0291 - accuracy: 0.9923 -
val_loss: 0.3014 - val_accuracy: 0.9205 - lr: 1.
0000e-05
Epoch 119/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0256 - accuracy: 0.9921
[Plateau Check] Variação últimas 30 épocas: 0.
004337
693/693 [==============================] - 198s
285ms/step - loss: 0.0256 - accuracy: 0.9921 -
val_loss: 0.3038 - val_accuracy: 0.9183 - lr: 1.
0000e-05
Epoch 120/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0265 - accuracy: 0.9915
[Plateau Check] Variação últimas 30 épocas: 0.
004337
693/693 [==============================] - 198s
285ms/step - loss: 0.0265 - accuracy: 0.9915 -
val_loss: 0.3022 - val_accuracy: 0.9198 - lr: 1.
0000e-05
Epoch 121/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0272 - accuracy: 0.9921
[Plateau Check] Variação últimas 30 épocas: 0.
004337
```

```
1116 693/693 [==============================] - 198s
     285ms/step - loss: 0.0272 - accuracy: 0.9921 -
     val_loss: 0.3026 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1117 Epoch 122/500
1118 693/693 [==============================] - ETA: 0s
      - loss: 0.0287 - accuracy: 0.9912
1119 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1120 693/693 [==============================] - 198s
     285ms/step - loss: 0.0287 - accuracy: 0.9912 -
     val_loss: 0.3020 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1121 Epoch 123/500
1122 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0261 - accuracy: 0.9926
1123 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1124 693/693 [==============================] - 198s
     285ms/step - loss: 0.0262 - accuracy: 0.9925 -
     val_loss: 0.3024 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1125 Epoch 124/500
1126 693/693 [==============================] - ETA: 0s
      - loss: 0.0259 - accuracy: 0.9930
1127 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1128 693/693 [==============================] - 198s
     285ms/step - loss: 0.0259 - accuracy: 0.9930 -
     val_loss: 0.3022 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1129 Epoch 125/500
1130 693/693 [==============================] - ETA: 0s
      - loss: 0.0251 - accuracy: 0.9935
1131 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1132 693/693 [==============================] - 198s
     285ms/step - loss: 0.0251 - accuracy: 0.9935 -
     val_loss: 0.3003 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1133 Epoch 126/500
```

```
1134 692/693 [============================>.] - ETA: 0s
      - loss: 0.0238 - accuracy: 0.9940
1135 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1136 693/693 [==============================] - 198s
     285ms/step - loss: 0.0239 - accuracy: 0.9939 -
     val_loss: 0.3018 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1137 Epoch 127/500
1138 693/693 [==============================] - ETA: 0s
      - loss: 0.0265 - accuracy: 0.9924
1139 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1140 693/693 [==============================] - 198s
     285ms/step - loss: 0.0265 - accuracy: 0.9924 -
     val_loss: 0.3033 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1141 Epoch 128/500
1142 692/693 [============================>.] - ETA: 0s
      - loss: 0.0241 - accuracy: 0.9945
1143 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1144 693/693 [==============================] - 198s
     285ms/step - loss: 0.0242 - accuracy: 0.9944 -
     val_loss: 0.3025 - val_accuracy: 0.9216 - lr: 1.
     0000e-05
1145 Epoch 129/500
1146 693/693 [==============================] - ETA: 0s
      - loss: 0.0272 - accuracy: 0.9920
1147 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1148 693/693 [==============================] - 198s
     285ms/step - loss: 0.0272 - accuracy: 0.9920 -
     val_loss: 0.3031 - val_accuracy: 0.9216 - lr: 1.
     0000e-05
1149 Epoch 130/500
1150 693/693 [==============================] - ETA: 0s
      - loss: 0.0266 - accuracy: 0.9913
1151 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1152 693/693 [==============================] - 198s
```

```
1152 285ms/step - loss: 0.0266 - accuracy: 0.9913 -
     val_loss: 0.3012 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1153 Epoch 131/500
1154 692/693 [============================>.] - ETA: 0s
      - loss: 0.0265 - accuracy: 0.9926
1155 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1156 693/693 [==============================] - 198s
     285ms/step - loss: 0.0270 - accuracy: 0.9925 -
     val_loss: 0.3027 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1157 Epoch 132/500
1158 692/693 [============================>.] - ETA: 0s
      - loss: 0.0277 - accuracy: 0.9916
1159 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1160 693/693 [==============================] - 198s
     285ms/step - loss: 0.0280 - accuracy: 0.9915 -
     val_loss: 0.3030 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1161 Epoch 133/500
1162 693/693 [==============================] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9918
1163 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1164 693/693 [==============================] - 198s
     285ms/step - loss: 0.0268 - accuracy: 0.9918 -
     val_loss: 0.3023 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1165 Epoch 134/500
1166 692/693 [============================>.] - ETA: 0s
      - loss: 0.0273 - accuracy: 0.9920
1167 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1168 693/693 [==============================] - 198s
     285ms/step - loss: 0.0273 - accuracy: 0.9920 -
     val_loss: 0.3030 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1169 Epoch 135/500
1170 692/693 [============================>.] - ETA: 0s
```

```
1170  - loss: 0.0250 - accuracy: 0.9934
1171 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1172 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9934 -
     val_loss: 0.3021 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1173 Epoch 136/500
1174 692/693 [============================>.] - ETA: 0s
      - loss: 0.0265 - accuracy: 0.9931
1175 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
1176 693/693 [==============================] - 199s
     286ms/step - loss: 0.0267 - accuracy: 0.9930 -
     val_loss: 0.3042 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1177 Epoch 137/500
1178 692/693 [============================>.] - ETA: 0s
      - loss: 0.0262 - accuracy: 0.9937
1179 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1180 693/693 [==============================] - 198s
     285ms/step - loss: 0.0268 - accuracy: 0.9936 -
     val_loss: 0.3021 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1181 Epoch 138/500
1182 693/693 [==============================] - ETA: 0s
      - loss: 0.0247 - accuracy: 0.9944
1183 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1184 693/693 [==============================] - 198s
     285ms/step - loss: 0.0247 - accuracy: 0.9944 -
     val_loss: 0.3018 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1185 Epoch 139/500
1186 692/693 [============================>.] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9923
1187 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1188 693/693 [==============================] - 198s
     285ms/step - loss: 0.0272 - accuracy: 0.9922 -
```

```
1188 val_loss: 0.3021 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1189 Epoch 140/500
1190 692/693 [============================>.] - ETA: 0s
      - loss: 0.0266 - accuracy: 0.9921
1191 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1192 693/693 [==============================] - 198s
     285ms/step - loss: 0.0266 - accuracy: 0.9921 -
     val_loss: 0.3001 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1193 Epoch 141/500
1194 692/693 [============================>.] - ETA: 0s
      - loss: 0.0273 - accuracy: 0.9924
1195 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1196 693/693 [==============================] - 198s
     285ms/step - loss: 0.0289 - accuracy: 0.9922 -
     val_loss: 0.3044 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1197 Epoch 142/500
1198 693/693 [==============================] - ETA: 0s
      - loss: 0.0292 - accuracy: 0.9909
1199 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1200 693/693 [==============================] - 198s
     285ms/step - loss: 0.0292 - accuracy: 0.9909 -
     val_loss: 0.3030 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1201 Epoch 143/500
1202 692/693 [============================>.] - ETA: 0s
      - loss: 0.0265 - accuracy: 0.9921
1203 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1204 693/693 [==============================] - 198s
     285ms/step - loss: 0.0271 - accuracy: 0.9921 -
     val_loss: 0.3041 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1205 Epoch 144/500
1206 692/693 [============================>.] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9921
```

```
1207 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1208 693/693 [==============================] - 198s
     285ms/step - loss: 0.0270 - accuracy: 0.9921 -
     val_loss: 0.3035 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1209 Epoch 145/500
1210 692/693 [============================>.] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9931
1211 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1212 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9931 -
     val_loss: 0.3025 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1213 Epoch 146/500
1214 693/693 [==============================] - ETA: 0s
      - loss: 0.0266 - accuracy: 0.9928
1215 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
1216 693/693 [==============================] - 198s
     285ms/step - loss: 0.0266 - accuracy: 0.9928 -
     val_loss: 0.3024 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1217 Epoch 147/500
1218 692/693 [============================>.] - ETA: 0s
      - loss: 0.0252 - accuracy: 0.9942
1219 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1220 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9941 -
     val_loss: 0.3034 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1221 Epoch 148/500
1222 692/693 [============================>.] - ETA: 0s
      - loss: 0.0264 - accuracy: 0.9924
1223 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1224 693/693 [==============================] - 198s
     285ms/step - loss: 0.0264 - accuracy: 0.9924 -
     val_loss: 0.3032 - val_accuracy: 0.9183 - lr: 1.
```

```
1224 0000e-05
1225 Epoch 149/500
1226 692/693 [============================>.] - ETA: 0s
      - loss: 0.0266 - accuracy: 0.9932
1227 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1228 693/693 [==============================] - 198s
     285ms/step - loss: 0.0266 - accuracy: 0.9932 -
     val_loss: 0.3024 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1229 Epoch 150/500
1230 692/693 [============================>.] - ETA: 0s
      - loss: 0.0278 - accuracy: 0.9923
1231 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1232 693/693 [==============================] - 198s
     285ms/step - loss: 0.0278 - accuracy: 0.9923 -
     val_loss: 0.3020 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1233 Epoch 151/500
1234 692/693 [============================>.] - ETA: 0s
      - loss: 0.0247 - accuracy: 0.9923
1235 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1236 693/693 [==============================] - 198s
     285ms/step - loss: 0.0248 - accuracy: 0.9923 -
     val_loss: 0.3038 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1237 Epoch 152/500
1238 692/693 [============================>.] - ETA: 0s
      - loss: 0.0264 - accuracy: 0.9931
1239 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1240 693/693 [==============================] - 198s
     285ms/step - loss: 0.0265 - accuracy: 0.9931 -
     val_loss: 0.3017 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1241 Epoch 153/500
1242 693/693 [==============================] - ETA: 0s
      - loss: 0.0277 - accuracy: 0.9923
1243 [Plateau Check] Variação últimas 30 épocas: 0.
```

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1243 003614
1244 693/693 [==============================] - 198s
     285ms/step - loss: 0.0277 - accuracy: 0.9923 -
     val_loss: 0.3029 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1245 Epoch 154/500
1246 692/693 [============================>.] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9923
1247 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1248 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9923 -
     val_loss: 0.3023 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1249 Epoch 155/500
1250 692/693 [============================>.] - ETA: 0s
      - loss: 0.0267 - accuracy: 0.9908
1251 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1252 693/693 [==============================] - 198s
     285ms/step - loss: 0.0269 - accuracy: 0.9907 -
     val_loss: 0.3026 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1253 Epoch 156/500
1254 692/693 [============================>.] - ETA: 0s
      - loss: 0.0272 - accuracy: 0.9927
1255 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1256 693/693 [==============================] - 198s
     285ms/step - loss: 0.0277 - accuracy: 0.9926 -
     val_loss: 0.3020 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1257 Epoch 157/500
1258 693/693 [==============================] - ETA: 0s
      - loss: 0.0235 - accuracy: 0.9941
1259 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1260 693/693 [==============================] - 198s
     285ms/step - loss: 0.0235 - accuracy: 0.9941 -
     val_loss: 0.3031 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
```

```
Epoch 158/500
692/693 [============================>.] - ETA: 0s - loss: 0.0280 - accuracy: 0.9914
[Plateau Check] Variação últimas 30 épocas: 0.003614
693/693 [==============================] - 198s 285ms/step - loss: 0.0282 - accuracy: 0.9913 - val_loss: 0.3018 - val_accuracy: 0.9205 - lr: 1.0000e-05
Epoch 159/500
693/693 [==============================] - ETA: 0s - loss: 0.0247 - accuracy: 0.9937
[Plateau Check] Variação últimas 30 épocas: 0.003253
693/693 [==============================] - 199s 285ms/step - loss: 0.0247 - accuracy: 0.9937 - val_loss: 0.3020 - val_accuracy: 0.9190 - lr: 1.0000e-05
Epoch 160/500
693/693 [==============================] - ETA: 0s - loss: 0.0283 - accuracy: 0.9912
[Plateau Check] Variação últimas 30 épocas: 0.003253
693/693 [==============================] - 198s 285ms/step - loss: 0.0283 - accuracy: 0.9912 - val_loss: 0.3020 - val_accuracy: 0.9190 - lr: 1.0000e-05
Epoch 161/500
692/693 [============================>.] - ETA: 0s - loss: 0.0242 - accuracy: 0.9934
[Plateau Check] Variação últimas 30 épocas: 0.003253
693/693 [==============================] - 198s 285ms/step - loss: 0.0247 - accuracy: 0.9933 - val_loss: 0.3015 - val_accuracy: 0.9212 - lr: 1.0000e-05
Epoch 162/500
693/693 [==============================] - ETA: 0s - loss: 0.0260 - accuracy: 0.9922
[Plateau Check] Variação últimas 30 épocas: 0.003253
```

```
693/693 [==============================] - 198s
    285ms/step - loss: 0.0260 - accuracy: 0.9922 -
    val_loss: 0.3030 - val_accuracy: 0.9201 - lr: 1.
    0000e-05
Epoch 163/500
693/693 [==============================] - ETA: 0s
     - loss: 0.0262 - accuracy: 0.9912
[Plateau Check] Variação últimas 30 épocas: 0.
    003253
693/693 [==============================] - 198s
    285ms/step - loss: 0.0262 - accuracy: 0.9912 -
    val_loss: 0.3021 - val_accuracy: 0.9194 - lr: 1.
    0000e-05
Epoch 164/500
692/693 [==============================>.] - ETA: 0s
     - loss: 0.0269 - accuracy: 0.9921
[Plateau Check] Variação últimas 30 épocas: 0.
    003253
693/693 [==============================] - 198s
    285ms/step - loss: 0.0273 - accuracy: 0.9921 -
    val_loss: 0.3044 - val_accuracy: 0.9209 - lr: 1.
    0000e-05
Epoch 165/500
693/693 [==============================] - ETA: 0s
     - loss: 0.0266 - accuracy: 0.9927
[Plateau Check] Variação últimas 30 épocas: 0.
    003253
693/693 [==============================] - 198s
    285ms/step - loss: 0.0266 - accuracy: 0.9927 -
    val_loss: 0.3029 - val_accuracy: 0.9187 - lr: 1.
    0000e-05
Epoch 166/500
692/693 [==============================>.] - ETA: 0s
     - loss: 0.0260 - accuracy: 0.9932
[Plateau Check] Variação últimas 30 épocas: 0.
    003253
693/693 [==============================] - 198s
    285ms/step - loss: 0.0262 - accuracy: 0.9931 -
    val_loss: 0.3035 - val_accuracy: 0.9201 - lr: 1.
    0000e-05
Epoch 167/500
```

```
1298 692/693 [============================>.] - ETA: 0s
      - loss: 0.0253 - accuracy: 0.9926
1299 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1300 693/693 [==============================] - 198s
     285ms/step - loss: 0.0253 - accuracy: 0.9926 -
     val_loss: 0.3030 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1301 Epoch 168/500
1302 692/693 [============================>.] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9934
1303 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1304 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9933 -
     val_loss: 0.3028 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1305 Epoch 169/500
1306 692/693 [============================>.] - ETA: 0s
      - loss: 0.0303 - accuracy: 0.9904
1307 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1308 693/693 [==============================] - 198s
     285ms/step - loss: 0.0306 - accuracy: 0.9903 -
     val_loss: 0.3035 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1309 Epoch 170/500
1310 692/693 [============================>.] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9928
1311 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1312 693/693 [==============================] - 198s
     285ms/step - loss: 0.0249 - accuracy: 0.9928 -
     val_loss: 0.3026 - val_accuracy: 0.9176 - lr: 1.
     0000e-05
1313 Epoch 171/500
1314 693/693 [==============================] - ETA: 0s
      - loss: 0.0252 - accuracy: 0.9931
1315 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1316 693/693 [==============================] - 198s
```

```
1316 285ms/step - loss: 0.0252 - accuracy: 0.9931 -
     val_loss: 0.3033 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1317 Epoch 172/500
1318 693/693 [==============================] - ETA: 0s
      - loss: 0.0251 - accuracy: 0.9933
1319 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1320 693/693 [==============================] - 198s
     285ms/step - loss: 0.0251 - accuracy: 0.9933 -
     val_loss: 0.3026 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1321 Epoch 173/500
1322 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0267 - accuracy: 0.9924
1323 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1324 693/693 [==============================] - 198s
     285ms/step - loss: 0.0267 - accuracy: 0.9924 -
     val_loss: 0.3038 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1325 Epoch 174/500
1326 693/693 [==============================] - ETA: 0s
      - loss: 0.0264 - accuracy: 0.9928
1327 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1328 693/693 [==============================] - 198s
     285ms/step - loss: 0.0264 - accuracy: 0.9928 -
     val_loss: 0.3037 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1329 Epoch 175/500
1330 693/693 [==============================] - ETA: 0s
      - loss: 0.0281 - accuracy: 0.9920
1331 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1332 693/693 [==============================] - 198s
     285ms/step - loss: 0.0281 - accuracy: 0.9920 -
     val_loss: 0.3047 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1333 Epoch 176/500
1334 692/693 [==============================>.] - ETA: 0s
```

```
1334  - loss: 0.0256 - accuracy: 0.9923
1335 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1336 693/693 [==============================] - 198s
     285ms/step - loss: 0.0256 - accuracy: 0.9923 -
     val_loss: 0.3037 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1337 Epoch 177/500
1338 693/693 [==============================] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9939
1339 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1340 693/693 [==============================] - 198s
     285ms/step - loss: 0.0244 - accuracy: 0.9939 -
     val_loss: 0.3041 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1341 Epoch 178/500
1342 692/693 [============================>.] - ETA: 0s
      - loss: 0.0273 - accuracy: 0.9923
1343 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1344 693/693 [==============================] - 198s
     285ms/step - loss: 0.0284 - accuracy: 0.9921 -
     val_loss: 0.3043 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1345 Epoch 179/500
1346 692/693 [============================>.] - ETA: 0s
      - loss: 0.0242 - accuracy: 0.9931
1347 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1348 693/693 [==============================] - 199s
     286ms/step - loss: 0.0242 - accuracy: 0.9931 -
     val_loss: 0.3037 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1349 Epoch 180/500
1350 692/693 [============================>.] - ETA: 0s
      - loss: 0.0267 - accuracy: 0.9926
1351 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1352 693/693 [==============================] - 199s
     285ms/step - loss: 0.0269 - accuracy: 0.9925 -
```

```
1352 val_loss: 0.3039 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1353 Epoch 181/500
1354 692/693 [============================>.] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9921
1355 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1356 693/693 [==============================] - 198s
     285ms/step - loss: 0.0268 - accuracy: 0.9921 -
     val_loss: 0.3041 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1357 Epoch 182/500
1358 692/693 [============================>.] - ETA: 0s
      - loss: 0.0254 - accuracy: 0.9931
1359 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1360 693/693 [==============================] - 198s
     285ms/step - loss: 0.0254 - accuracy: 0.9931 -
     val_loss: 0.3033 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1361 Epoch 183/500
1362 692/693 [============================>.] - ETA: 0s
      - loss: 0.0266 - accuracy: 0.9928
1363 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1364 693/693 [==============================] - 198s
     285ms/step - loss: 0.0269 - accuracy: 0.9927 -
     val_loss: 0.3043 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1365 Epoch 184/500
1366 693/693 [==============================] - ETA: 0s
      - loss: 0.0290 - accuracy: 0.9916
1367 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1368 693/693 [==============================] - 198s
     285ms/step - loss: 0.0290 - accuracy: 0.9916 -
     val_loss: 0.3045 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1369 Epoch 185/500
1370 692/693 [============================>.] - ETA: 0s
      - loss: 0.0243 - accuracy: 0.9935
```

```
1371 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1372 693/693 [==============================] - 198s
     285ms/step - loss: 0.0245 - accuracy: 0.9934 -
     val_loss: 0.3044 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1373 Epoch 186/500
1374 692/693 [============================>.] - ETA: 0s
      - loss: 0.0270 - accuracy: 0.9921
1375 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1376 693/693 [==============================] - 198s
     285ms/step - loss: 0.0275 - accuracy: 0.9921 -
     val_loss: 0.3041 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1377 Epoch 187/500
1378 692/693 [============================>.] - ETA: 0s
      - loss: 0.0256 - accuracy: 0.9926
1379 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1380 693/693 [==============================] - 198s
     285ms/step - loss: 0.0263 - accuracy: 0.9924 -
     val_loss: 0.3036 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1381 Epoch 188/500
1382 692/693 [============================>.] - ETA: 0s
      - loss: 0.0260 - accuracy: 0.9929
1383 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1384 693/693 [==============================] - 198s
     285ms/step - loss: 0.0265 - accuracy: 0.9928 -
     val_loss: 0.3050 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1385 Epoch 189/500
1386 693/693 [==============================] - ETA: 0s
      - loss: 0.0255 - accuracy: 0.9924
1387 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1388 693/693 [==============================] - 198s
     285ms/step - loss: 0.0255 - accuracy: 0.9924 -
     val_loss: 0.3039 - val_accuracy: 0.9205 - lr: 1.
```

```
1388 0000e-05
1389 Epoch 190/500
1390 693/693 [==============================] - ETA: 0s
      - loss: 0.0294 - accuracy: 0.9916
1391 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1392 693/693 [==============================] - 199s
     285ms/step - loss: 0.0294 - accuracy: 0.9916 -
     val_loss: 0.3056 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1393 Epoch 191/500
1394 693/693 [==============================] - ETA: 0s
      - loss: 0.0255 - accuracy: 0.9931
1395 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1396 693/693 [==============================] - 198s
     285ms/step - loss: 0.0255 - accuracy: 0.9931 -
     val_loss: 0.3041 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1397 Epoch 192/500
1398 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0276 - accuracy: 0.9929
1399 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1400 693/693 [==============================] - 198s
     285ms/step - loss: 0.0276 - accuracy: 0.9929 -
     val_loss: 0.3045 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1401 Epoch 193/500
1402 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0255 - accuracy: 0.9930
1403 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1404 693/693 [==============================] - 198s
     285ms/step - loss: 0.0260 - accuracy: 0.9929 -
     val_loss: 0.3045 - val_accuracy: 0.9216 - lr: 1.
     0000e-05
1405 Epoch 194/500
1406 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0269 - accuracy: 0.9925
1407 [Plateau Check] Variação últimas 30 épocas: 0.
```

```
003975
693/693 [==============================] - 198s
    285ms/step - loss: 0.0271 - accuracy: 0.9923 -
    val_loss: 0.3049 - val_accuracy: 0.9209 - lr: 1.
    0000e-05
Epoch 195/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0254 - accuracy: 0.9931
[Plateau Check] Variação últimas 30 épocas: 0.
    003975
693/693 [==============================] - 198s
    285ms/step - loss: 0.0254 - accuracy: 0.9931 -
    val_loss: 0.3039 - val_accuracy: 0.9205 - lr: 1.
    0000e-05
Epoch 196/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0260 - accuracy: 0.9930
[Plateau Check] Variação últimas 30 épocas: 0.
    003975
693/693 [==============================] - 198s
    285ms/step - loss: 0.0260 - accuracy: 0.9930 -
    val_loss: 0.3037 - val_accuracy: 0.9201 - lr: 1.
    0000e-05
Epoch 197/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0256 - accuracy: 0.9921
[Plateau Check] Variação últimas 30 épocas: 0.
    003975
693/693 [==============================] - 198s
    285ms/step - loss: 0.0261 - accuracy: 0.9921 -
    val_loss: 0.3033 - val_accuracy: 0.9201 - lr: 1.
    0000e-05
Epoch 198/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0269 - accuracy: 0.9928
[Plateau Check] Variação últimas 30 épocas: 0.
    003975
693/693 [==============================] - 198s
    285ms/step - loss: 0.0273 - accuracy: 0.9927 -
    val_loss: 0.3039 - val_accuracy: 0.9198 - lr: 1.
    0000e-05
```

```
Epoch 199/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0237 - accuracy: 0.9933
[Plateau Check] Variação últimas 30 épocas: 0.
003975
693/693 [==============================] - 198s
285ms/step - loss: 0.0238 - accuracy: 0.9932 -
val_loss: 0.3038 - val_accuracy: 0.9187 - lr: 1.
0000e-05
Epoch 200/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0229 - accuracy: 0.9946
[Plateau Check] Variação últimas 30 épocas: 0.
003614
693/693 [==============================] - 198s
285ms/step - loss: 0.0232 - accuracy: 0.9945 -
val_loss: 0.3031 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 201/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0260 - accuracy: 0.9923
[Plateau Check] Variação últimas 30 épocas: 0.
003614
693/693 [==============================] - 198s
285ms/step - loss: 0.0260 - accuracy: 0.9923 -
val_loss: 0.3039 - val_accuracy: 0.9205 - lr: 1.
0000e-05
Epoch 202/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0270 - accuracy: 0.9914
[Plateau Check] Variação últimas 30 épocas: 0.
003614
693/693 [==============================] - 198s
285ms/step - loss: 0.0270 - accuracy: 0.9914 -
val_loss: 0.3044 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 203/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0268 - accuracy: 0.9922
[Plateau Check] Variação últimas 30 épocas: 0.
003614
```

```
1444 693/693 [==============================] - 198s
     285ms/step - loss: 0.0268 - accuracy: 0.9922 -
     val_loss: 0.3036 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1445 Epoch 204/500
1446 692/693 [============================>.] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9932
1447 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1448 693/693 [==============================] - 198s
     285ms/step - loss: 0.0245 - accuracy: 0.9931 -
     val_loss: 0.3051 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1449 Epoch 205/500
1450 692/693 [============================>.] - ETA: 0s
      - loss: 0.0279 - accuracy: 0.9917
1451 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1452 693/693 [==============================] - 198s
     285ms/step - loss: 0.0282 - accuracy: 0.9916 -
     val_loss: 0.3050 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1453 Epoch 206/500
1454 692/693 [============================>.] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9925
1455 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1456 693/693 [==============================] - 198s
     285ms/step - loss: 0.0259 - accuracy: 0.9924 -
     val_loss: 0.3049 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1457 Epoch 207/500
1458 692/693 [============================>.] - ETA: 0s
      - loss: 0.0263 - accuracy: 0.9914
1459 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1460 693/693 [==============================] - 198s
     285ms/step - loss: 0.0268 - accuracy: 0.9913 -
     val_loss: 0.3039 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1461 Epoch 208/500
```

```
1462 693/693 [==============================] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9925
1463 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1464 693/693 [==============================] - 198s
     285ms/step - loss: 0.0268 - accuracy: 0.9925 -
     val_loss: 0.3046 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1465 Epoch 209/500
1466 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9930
1467 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1468 693/693 [==============================] - 199s
     286ms/step - loss: 0.0257 - accuracy: 0.9929 -
     val_loss: 0.3047 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1469 Epoch 210/500
1470 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9921
1471 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1472 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9921 -
     val_loss: 0.3033 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1473 Epoch 211/500
1474 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0288 - accuracy: 0.9917
1475 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1476 693/693 [==============================] - 198s
     285ms/step - loss: 0.0288 - accuracy: 0.9917 -
     val_loss: 0.3045 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1477 Epoch 212/500
1478 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0270 - accuracy: 0.9917
1479 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1480 693/693 [==============================] - 198s
```

```
1480 285ms/step - loss: 0.0270 - accuracy: 0.9917 -
     val_loss: 0.3055 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1481 Epoch 213/500
1482 692/693 [============================>.] - ETA: 0s
      - loss: 0.0283 - accuracy: 0.9925
1483 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1484 693/693 [==============================] - 198s
     285ms/step - loss: 0.0283 - accuracy: 0.9925 -
     val_loss: 0.3041 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1485 Epoch 214/500
1486 693/693 [==============================] - ETA: 0s
      - loss: 0.0260 - accuracy: 0.9925
1487 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1488 693/693 [==============================] - 198s
     285ms/step - loss: 0.0260 - accuracy: 0.9925 -
     val_loss: 0.3041 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1489 Epoch 215/500
1490 692/693 [============================>.] - ETA: 0s
      - loss: 0.0240 - accuracy: 0.9938
1491 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1492 693/693 [==============================] - 198s
     285ms/step - loss: 0.0244 - accuracy: 0.9937 -
     val_loss: 0.3049 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1493 Epoch 216/500
1494 693/693 [==============================] - ETA: 0s
      - loss: 0.0256 - accuracy: 0.9918
1495 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1496 693/693 [==============================] - 198s
     285ms/step - loss: 0.0256 - accuracy: 0.9918 -
     val_loss: 0.3031 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1497 Epoch 217/500
1498 693/693 [==============================] - ETA: 0s
```

```
1498  - loss: 0.0236 - accuracy: 0.9927
1499 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1500 693/693 [==============================] - 198s
     285ms/step - loss: 0.0236 - accuracy: 0.9927 -
     val_loss: 0.3042 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1501 Epoch 218/500
1502 693/693 [==============================] - ETA: 0s
      - loss: 0.0215 - accuracy: 0.9944
1503 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1504 693/693 [==============================] - 198s
     285ms/step - loss: 0.0215 - accuracy: 0.9944 -
     val_loss: 0.3027 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1505 Epoch 219/500
1506 692/693 [============================>.] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9923
1507 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1508 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9923 -
     val_loss: 0.3034 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1509 Epoch 220/500
1510 692/693 [============================>.] - ETA: 0s
      - loss: 0.0251 - accuracy: 0.9933
1511 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1512 693/693 [==============================] - 198s
     285ms/step - loss: 0.0251 - accuracy: 0.9933 -
     val_loss: 0.3040 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1513 Epoch 221/500
1514 693/693 [==============================] - ETA: 0s
      - loss: 0.0269 - accuracy: 0.9937
1515 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1516 693/693 [==============================] - 199s
     286ms/step - loss: 0.0269 - accuracy: 0.9937 -
```

```
1516 val_loss: 0.3041 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1517 Epoch 222/500
1518 692/693 [============================>.] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9921
1519 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1520 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9920 -
     val_loss: 0.3054 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1521 Epoch 223/500
1522 692/693 [============================>.] - ETA: 0s
      - loss: 0.0248 - accuracy: 0.9925
1523 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1524 693/693 [==============================] - 198s
     285ms/step - loss: 0.0248 - accuracy: 0.9925 -
     val_loss: 0.3049 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1525 Epoch 224/500
1526 692/693 [============================>.] - ETA: 0s
      - loss: 0.0243 - accuracy: 0.9936
1527 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1528 693/693 [==============================] - 198s
     285ms/step - loss: 0.0244 - accuracy: 0.9936 -
     val_loss: 0.3045 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1529 Epoch 225/500
1530 692/693 [============================>.] - ETA: 0s
      - loss: 0.0238 - accuracy: 0.9936
1531 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1532 693/693 [==============================] - 198s
     285ms/step - loss: 0.0238 - accuracy: 0.9936 -
     val_loss: 0.3048 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1533 Epoch 226/500
1534 692/693 [============================>.] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9921
```

```
1535 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1536 693/693 [==============================] - 198s
     285ms/step - loss: 0.0273 - accuracy: 0.9921 -
     val_loss: 0.3048 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1537 Epoch 227/500
1538 693/693 [==============================] - ETA: 0s
      - loss: 0.0252 - accuracy: 0.9926
1539 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1540 693/693 [==============================] - 198s
     285ms/step - loss: 0.0252 - accuracy: 0.9926 -
     val_loss: 0.3042 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1541 Epoch 228/500
1542 693/693 [==============================] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9938
1543 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1544 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9938 -
     val_loss: 0.3049 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1545 Epoch 229/500
1546 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9932
1547 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1548 693/693 [==============================] - 198s
     285ms/step - loss: 0.0243 - accuracy: 0.9931 -
     val_loss: 0.3042 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1549 Epoch 230/500
1550 693/693 [==============================] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9932
1551 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1552 693/693 [==============================] - 198s
     285ms/step - loss: 0.0237 - accuracy: 0.9932 -
     val_loss: 0.3039 - val_accuracy: 0.9198 - lr: 1.
```

```
1552 0000e-05
1553 Epoch 231/500
1554 693/693 [==============================] - ETA: 0s
      - loss: 0.0269 - accuracy: 0.9925
1555 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1556 693/693 [==============================] - 198s
     285ms/step - loss: 0.0269 - accuracy: 0.9925 -
     val_loss: 0.3038 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1557 Epoch 232/500
1558 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0246 - accuracy: 0.9930
1559 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1560 693/693 [==============================] - 198s
     285ms/step - loss: 0.0249 - accuracy: 0.9928 -
     val_loss: 0.3063 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1561 Epoch 233/500
1562 693/693 [==============================] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9937
1563 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1564 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9937 -
     val_loss: 0.3044 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1565 Epoch 234/500
1566 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0219 - accuracy: 0.9946
1567 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1568 693/693 [==============================] - 198s
     285ms/step - loss: 0.0220 - accuracy: 0.9946 -
     val_loss: 0.3048 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1569 Epoch 235/500
1570 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0256 - accuracy: 0.9930
1571 [Plateau Check] Variação últimas 30 épocas: 0.
```

```
002891
693/693 [==============================] - 198s
    285ms/step - loss: 0.0256 - accuracy: 0.9930 -
    val_loss: 0.3053 - val_accuracy: 0.9190 - lr: 1.
    0000e-05
Epoch 236/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0255 - accuracy: 0.9930
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 198s
    285ms/step - loss: 0.0254 - accuracy: 0.9930 -
    val_loss: 0.3038 - val_accuracy: 0.9194 - lr: 1.
    0000e-05
Epoch 237/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0262 - accuracy: 0.9925
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 198s
    285ms/step - loss: 0.0266 - accuracy: 0.9924 -
    val_loss: 0.3052 - val_accuracy: 0.9209 - lr: 1.
    0000e-05
Epoch 238/500
693/693 [==============================] - ETA: 0s
     - loss: 0.0238 - accuracy: 0.9934
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 198s
    285ms/step - loss: 0.0238 - accuracy: 0.9934 -
    val_loss: 0.3041 - val_accuracy: 0.9209 - lr: 1.
    0000e-05
Epoch 239/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0255 - accuracy: 0.9931
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 198s
    285ms/step - loss: 0.0255 - accuracy: 0.9931 -
    val_loss: 0.3050 - val_accuracy: 0.9198 - lr: 1.
    0000e-05
```

```
Epoch 240/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0229 - accuracy: 0.9946
[Plateau Check] Variação últimas 30 épocas: 0.
002891
693/693 [==============================] - 198s
285ms/step - loss: 0.0229 - accuracy: 0.9946 -
val_loss: 0.3026 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 241/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0263 - accuracy: 0.9921
[Plateau Check] Variação últimas 30 épocas: 0.
002891
693/693 [==============================] - 198s
285ms/step - loss: 0.0263 - accuracy: 0.9921 -
val_loss: 0.3032 - val_accuracy: 0.9194 - lr: 1.
0000e-05
Epoch 242/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0248 - accuracy: 0.9929
[Plateau Check] Variação últimas 30 épocas: 0.
002891
693/693 [==============================] - 198s
285ms/step - loss: 0.0248 - accuracy: 0.9929 -
val_loss: 0.3025 - val_accuracy: 0.9187 - lr: 1.
0000e-05
Epoch 243/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0244 - accuracy: 0.9939
[Plateau Check] Variação últimas 30 épocas: 0.
002891
693/693 [==============================] - 198s
285ms/step - loss: 0.0244 - accuracy: 0.9939 -
val_loss: 0.3041 - val_accuracy: 0.9212 - lr: 1.
0000e-05
Epoch 244/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0264 - accuracy: 0.9913
[Plateau Check] Variação últimas 30 épocas: 0.
002891
```

```
1608 693/693 [==============================] - 198s
     285ms/step - loss: 0.0264 - accuracy: 0.9913 -
     val_loss: 0.3040 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1609 Epoch 245/500
1610 693/693 [==============================] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9936
1611 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1612 693/693 [==============================] - 198s
     285ms/step - loss: 0.0249 - accuracy: 0.9936 -
     val_loss: 0.3044 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1613 Epoch 246/500
1614 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0254 - accuracy: 0.9925
1615 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1616 693/693 [==============================] - 198s
     285ms/step - loss: 0.0254 - accuracy: 0.9925 -
     val_loss: 0.3046 - val_accuracy: 0.9219 - lr: 1.
     0000e-05
1617 Epoch 247/500
1618 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0264 - accuracy: 0.9928
1619 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1620 693/693 [==============================] - 198s
     285ms/step - loss: 0.0264 - accuracy: 0.9928 -
     val_loss: 0.3055 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1621 Epoch 248/500
1622 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0283 - accuracy: 0.9917
1623 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1624 693/693 [==============================] - 198s
     285ms/step - loss: 0.0286 - accuracy: 0.9916 -
     val_loss: 0.3052 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1625 Epoch 249/500
```

```
1626 692/693 [============================>.] - ETA: 0s
      - loss: 0.0263 - accuracy: 0.9929
1627 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1628 693/693 [==============================] - 198s
     285ms/step - loss: 0.0263 - accuracy: 0.9929 -
     val_loss: 0.3046 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1629 Epoch 250/500
1630 692/693 [============================>.] - ETA: 0s
      - loss: 0.0268 - accuracy: 0.9923
1631 [Plateau Check] Variação últimas 30 épocas: 0.
     003253
1632 693/693 [==============================] - 198s
     285ms/step - loss: 0.0270 - accuracy: 0.9922 -
     val_loss: 0.3043 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1633 Epoch 251/500
1634 692/693 [============================>.] - ETA: 0s
      - loss: 0.0259 - accuracy: 0.9928
1635 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1636 693/693 [==============================] - 198s
     285ms/step - loss: 0.0259 - accuracy: 0.9928 -
     val_loss: 0.3060 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1637 Epoch 252/500
1638 692/693 [============================>.] - ETA: 0s
      - loss: 0.0260 - accuracy: 0.9930
1639 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1640 693/693 [==============================] - 198s
     285ms/step - loss: 0.0260 - accuracy: 0.9930 -
     val_loss: 0.3042 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1641 Epoch 253/500
1642 692/693 [============================>.] - ETA: 0s
      - loss: 0.0240 - accuracy: 0.9935
1643 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1644 693/693 [==============================] - 198s
```

```
1644 285ms/step - loss: 0.0241 - accuracy: 0.9934 -
     val_loss: 0.3048 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1645 Epoch 254/500
1646 692/693 [============================>.] - ETA: 0s
      - loss: 0.0242 - accuracy: 0.9939
1647 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1648 693/693 [==============================] - 198s
     285ms/step - loss: 0.0243 - accuracy: 0.9939 -
     val_loss: 0.3049 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1649 Epoch 255/500
1650 692/693 [============================>.] - ETA: 0s
      - loss: 0.0259 - accuracy: 0.9923
1651 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1652 693/693 [==============================] - 198s
     285ms/step - loss: 0.0259 - accuracy: 0.9923 -
     val_loss: 0.3054 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1653 Epoch 256/500
1654 693/693 [==============================] - ETA: 0s
      - loss: 0.0252 - accuracy: 0.9935
1655 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1656 693/693 [==============================] - 198s
     285ms/step - loss: 0.0252 - accuracy: 0.9935 -
     val_loss: 0.3041 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1657 Epoch 257/500
1658 692/693 [============================>.] - ETA: 0s
      - loss: 0.0265 - accuracy: 0.9924
1659 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1660 693/693 [==============================] - 198s
     285ms/step - loss: 0.0265 - accuracy: 0.9924 -
     val_loss: 0.3032 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1661 Epoch 258/500
1662 693/693 [==============================] - ETA: 0s
```

```
1662  - loss: 0.0237 - accuracy: 0.9939
1663 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1664 693/693 [==============================] - 198s
     285ms/step - loss: 0.0237 - accuracy: 0.9939 -
     val_loss: 0.3050 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1665 Epoch 259/500
1666 692/693 [============================>.] - ETA: 0s
      - loss: 0.0247 - accuracy: 0.9924
1667 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1668 693/693 [==============================] - 198s
     285ms/step - loss: 0.0247 - accuracy: 0.9924 -
     val_loss: 0.3046 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1669 Epoch 260/500
1670 692/693 [============================>.] - ETA: 0s
      - loss: 0.0246 - accuracy: 0.9933
1671 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1672 693/693 [==============================] - 198s
     285ms/step - loss: 0.0247 - accuracy: 0.9933 -
     val_loss: 0.3042 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1673 Epoch 261/500
1674 693/693 [==============================] - ETA: 0s
      - loss: 0.0216 - accuracy: 0.9939
1675 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1676 693/693 [==============================] - 198s
     285ms/step - loss: 0.0216 - accuracy: 0.9939 -
     val_loss: 0.3045 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1677 Epoch 262/500
1678 692/693 [============================>.] - ETA: 0s
      - loss: 0.0239 - accuracy: 0.9933
1679 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1680 693/693 [==============================] - 198s
     285ms/step - loss: 0.0239 - accuracy: 0.9933 -
```

```
1680 val_loss: 0.3047 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1681 Epoch 263/500
1682 693/693 [==============================] - ETA: 0s
      - loss: 0.0234 - accuracy: 0.9943
1683 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1684 693/693 [==============================] - 198s
     285ms/step - loss: 0.0234 - accuracy: 0.9943 -
     val_loss: 0.3052 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1685 Epoch 264/500
1686 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0262 - accuracy: 0.9916
1687 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1688 693/693 [==============================] - 198s
     285ms/step - loss: 0.0263 - accuracy: 0.9916 -
     val_loss: 0.3050 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1689 Epoch 265/500
1690 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0253 - accuracy: 0.9934
1691 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1692 693/693 [==============================] - 198s
     285ms/step - loss: 0.0253 - accuracy: 0.9934 -
     val_loss: 0.3044 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1693 Epoch 266/500
1694 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0240 - accuracy: 0.9926
1695 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1696 693/693 [==============================] - 198s
     285ms/step - loss: 0.0240 - accuracy: 0.9926 -
     val_loss: 0.3048 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1697 Epoch 267/500
1698 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0231 - accuracy: 0.9939
```

```
1699 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1700 693/693 [==============================] - 198s
     285ms/step - loss: 0.0234 - accuracy: 0.9939 -
     val_loss: 0.3054 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1701 Epoch 268/500
1702 693/693 [==============================] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9948
1703 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1704 693/693 [==============================] - 198s
     285ms/step - loss: 0.0236 - accuracy: 0.9948 -
     val_loss: 0.3049 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1705 Epoch 269/500
1706 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0231 - accuracy: 0.9939
1707 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1708 693/693 [==============================] - 198s
     285ms/step - loss: 0.0234 - accuracy: 0.9939 -
     val_loss: 0.3046 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1709 Epoch 270/500
1710 693/693 [==============================] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9930
1711 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1712 693/693 [==============================] - 198s
     285ms/step - loss: 0.0249 - accuracy: 0.9930 -
     val_loss: 0.3042 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1713 Epoch 271/500
1714 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9923
1715 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1716 693/693 [==============================] - 198s
     285ms/step - loss: 0.0249 - accuracy: 0.9923 -
     val_loss: 0.3059 - val_accuracy: 0.9198 - lr: 1.
```

```
1716 0000e-05
1717 Epoch 272/500
1718 692/693 [============================>.] - ETA: 0s
      - loss: 0.0271 - accuracy: 0.9923
1719 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1720 693/693 [==============================] - 198s
     285ms/step - loss: 0.0271 - accuracy: 0.9923 -
     val_loss: 0.3047 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1721 Epoch 273/500
1722 692/693 [============================>.] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9945
1723 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1724 693/693 [==============================] - 198s
     285ms/step - loss: 0.0230 - accuracy: 0.9945 -
     val_loss: 0.3050 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1725 Epoch 274/500
1726 692/693 [============================>.] - ETA: 0s
      - loss: 0.0242 - accuracy: 0.9934
1727 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1728 693/693 [==============================] - 198s
     285ms/step - loss: 0.0242 - accuracy: 0.9934 -
     val_loss: 0.3047 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1729 Epoch 275/500
1730 692/693 [============================>.] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9930
1731 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1732 693/693 [==============================] - 198s
     285ms/step - loss: 0.0237 - accuracy: 0.9930 -
     val_loss: 0.3037 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1733 Epoch 276/500
1734 692/693 [============================>.] - ETA: 0s
      - loss: 0.0227 - accuracy: 0.9938
1735 [Plateau Check] Variação últimas 30 épocas: 0.
```

```
1735 002891
1736 693/693 [==============================] - 198s
     285ms/step - loss: 0.0232 - accuracy: 0.9936 -
     val_loss: 0.3043 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1737 Epoch 277/500
1738 692/693 [============================>.] - ETA: 0s
      - loss: 0.0255 - accuracy: 0.9926
1739 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1740 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9924 -
     val_loss: 0.3050 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1741 Epoch 278/500
1742 692/693 [============================>.] - ETA: 0s
      - loss: 0.0248 - accuracy: 0.9925
1743 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1744 693/693 [==============================] - 198s
     285ms/step - loss: 0.0248 - accuracy: 0.9925 -
     val_loss: 0.3052 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1745 Epoch 279/500
1746 692/693 [============================>.] - ETA: 0s
      - loss: 0.0271 - accuracy: 0.9934
1747 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1748 693/693 [==============================] - 198s
     285ms/step - loss: 0.0275 - accuracy: 0.9933 -
     val_loss: 0.3050 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1749 Epoch 280/500
1750 692/693 [============================>.] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9939
1751 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1752 693/693 [==============================] - 198s
     285ms/step - loss: 0.0241 - accuracy: 0.9939 -
     val_loss: 0.3060 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
```

```
Epoch 281/500
692/693 [============================>.] - ETA: 0s - loss: 0.0276 - accuracy: 0.9922
[Plateau Check] Variação últimas 30 épocas: 0.002530
693/693 [==============================] - 198s 285ms/step - loss: 0.0276 - accuracy: 0.9922 - val_loss: 0.3039 - val_accuracy: 0.9194 - lr: 1.0000e-05
Epoch 282/500
692/693 [============================>.] - ETA: 0s - loss: 0.0236 - accuracy: 0.9938
[Plateau Check] Variação últimas 30 épocas: 0.002530
693/693 [==============================] - 198s 285ms/step - loss: 0.0238 - accuracy: 0.9937 - val_loss: 0.3049 - val_accuracy: 0.9198 - lr: 1.0000e-05
Epoch 283/500
692/693 [============================>.] - ETA: 0s - loss: 0.0274 - accuracy: 0.9932
[Plateau Check] Variação últimas 30 épocas: 0.002530
693/693 [==============================] - 198s 285ms/step - loss: 0.0275 - accuracy: 0.9931 - val_loss: 0.3064 - val_accuracy: 0.9201 - lr: 1.0000e-05
Epoch 284/500
692/693 [============================>.] - ETA: 0s - loss: 0.0250 - accuracy: 0.9915
[Plateau Check] Variação últimas 30 épocas: 0.002168
693/693 [==============================] - 198s 285ms/step - loss: 0.0252 - accuracy: 0.9914 - val_loss: 0.3062 - val_accuracy: 0.9198 - lr: 1.0000e-05
Epoch 285/500
692/693 [============================>.] - ETA: 0s - loss: 0.0256 - accuracy: 0.9922
[Plateau Check] Variação últimas 30 épocas: 0.002168
```

```
1772 693/693 [==============================] - 198s
     285ms/step - loss: 0.0256 - accuracy: 0.9922 -
     val_loss: 0.3042 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1773 Epoch 286/500
1774 693/693 [==============================] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9931
1775 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
1776 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9931 -
     val_loss: 0.3062 - val_accuracy: 0.9209 - lr: 1.
     0000e-05
1777 Epoch 287/500
1778 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0263 - accuracy: 0.9918
1779 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
1780 693/693 [==============================] - 198s
     285ms/step - loss: 0.0263 - accuracy: 0.9918 -
     val_loss: 0.3040 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1781 Epoch 288/500
1782 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0218 - accuracy: 0.9950
1783 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
1784 693/693 [==============================] - 198s
     285ms/step - loss: 0.0222 - accuracy: 0.9949 -
     val_loss: 0.3054 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1785 Epoch 289/500
1786 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0210 - accuracy: 0.9946
1787 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1788 693/693 [==============================] - 198s
     285ms/step - loss: 0.0213 - accuracy: 0.9945 -
     val_loss: 0.3051 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1789 Epoch 290/500
```

```
1790 692/693 [============================>.] - ETA: 0s
      - loss: 0.0271 - accuracy: 0.9921
1791 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1792 693/693 [==============================] - 198s
     285ms/step - loss: 0.0272 - accuracy: 0.9921 -
     val_loss: 0.3058 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1793 Epoch 291/500
1794 692/693 [============================>.] - ETA: 0s
      - loss: 0.0248 - accuracy: 0.9926
1795 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1796 693/693 [==============================] - 198s
     285ms/step - loss: 0.0248 - accuracy: 0.9926 -
     val_loss: 0.3063 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1797 Epoch 292/500
1798 693/693 [==============================] - ETA: 0s
      - loss: 0.0274 - accuracy: 0.9926
1799 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1800 693/693 [==============================] - 198s
     285ms/step - loss: 0.0274 - accuracy: 0.9926 -
     val_loss: 0.3062 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
1801 Epoch 293/500
1802 692/693 [============================>.] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9941
1803 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1804 693/693 [==============================] - 198s
     285ms/step - loss: 0.0229 - accuracy: 0.9940 -
     val_loss: 0.3051 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1805 Epoch 294/500
1806 692/693 [============================>.] - ETA: 0s
      - loss: 0.0267 - accuracy: 0.9922
1807 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
1808 693/693 [==============================] - 199s
```

```
1808 285ms/step - loss: 0.0267 - accuracy: 0.9922 -
     val_loss: 0.3066 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1809 Epoch 295/500
1810 693/693 [==============================] - ETA: 0s
      - loss: 0.0220 - accuracy: 0.9941
1811 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1812 693/693 [==============================] - 198s
     285ms/step - loss: 0.0220 - accuracy: 0.9941 -
     val_loss: 0.3067 - val_accuracy: 0.9176 - lr: 1.
     0000e-05
1813 Epoch 296/500
1814 693/693 [==============================] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9925
1815 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1816 693/693 [==============================] - 198s
     285ms/step - loss: 0.0258 - accuracy: 0.9925 -
     val_loss: 0.3068 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1817 Epoch 297/500
1818 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0234 - accuracy: 0.9938
1819 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1820 693/693 [==============================] - 198s
     285ms/step - loss: 0.0234 - accuracy: 0.9938 -
     val_loss: 0.3051 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1821 Epoch 298/500
1822 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0259 - accuracy: 0.9922
1823 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1824 693/693 [==============================] - 198s
     285ms/step - loss: 0.0263 - accuracy: 0.9921 -
     val_loss: 0.3058 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1825 Epoch 299/500
1826 692/693 [==============================>.] - ETA: 0s
```

```
1826  - loss: 0.0227 - accuracy: 0.9943
1827 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1828 693/693 [==============================] - 198s
     285ms/step - loss: 0.0228 - accuracy: 0.9942 -
     val_loss: 0.3053 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1829 Epoch 300/500
1830 693/693 [==============================] - ETA: 0s
      - loss: 0.0240 - accuracy: 0.9936
1831 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1832 693/693 [==============================] - 198s
     285ms/step - loss: 0.0240 - accuracy: 0.9936 -
     val_loss: 0.3068 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1833 Epoch 301/500
1834 693/693 [==============================] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9936
1835 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1836 693/693 [==============================] - 198s
     285ms/step - loss: 0.0249 - accuracy: 0.9936 -
     val_loss: 0.3058 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1837 Epoch 302/500
1838 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0239 - accuracy: 0.9940
1839 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1840 693/693 [==============================] - 198s
     285ms/step - loss: 0.0240 - accuracy: 0.9940 -
     val_loss: 0.3063 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1841 Epoch 303/500
1842 693/693 [==============================] - ETA: 0s
      - loss: 0.0246 - accuracy: 0.9935
1843 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1844 693/693 [==============================] - 198s
     285ms/step - loss: 0.0246 - accuracy: 0.9935 -
```

```
1844 val_loss: 0.3077 - val_accuracy: 0.9172 - lr: 1.
     0000e-05
1845 Epoch 304/500
1846 692/693 [============================>.] - ETA: 0s
      - loss: 0.0220 - accuracy: 0.9949
1847 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1848 693/693 [==============================] - 198s
     285ms/step - loss: 0.0223 - accuracy: 0.9948 -
     val_loss: 0.3066 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1849 Epoch 305/500
1850 692/693 [============================>.] - ETA: 0s
      - loss: 0.0246 - accuracy: 0.9933
1851 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1852 693/693 [==============================] - 198s
     285ms/step - loss: 0.0247 - accuracy: 0.9933 -
     val_loss: 0.3066 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1853 Epoch 306/500
1854 693/693 [==============================] - ETA: 0s
      - loss: 0.0265 - accuracy: 0.9921
1855 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1856 693/693 [==============================] - 198s
     285ms/step - loss: 0.0265 - accuracy: 0.9921 -
     val_loss: 0.3067 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1857 Epoch 307/500
1858 692/693 [============================>.] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9930
1859 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1860 693/693 [==============================] - 198s
     285ms/step - loss: 0.0261 - accuracy: 0.9929 -
     val_loss: 0.3068 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1861 Epoch 308/500
1862 692/693 [============================>.] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9938
```

```
1863 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1864 693/693 [==============================] - 198s
     285ms/step - loss: 0.0230 - accuracy: 0.9938 -
     val_loss: 0.3075 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1865 Epoch 309/500
1866 692/693 [============================>.] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9938
1867 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1868 693/693 [==============================] - 198s
     285ms/step - loss: 0.0239 - accuracy: 0.9937 -
     val_loss: 0.3070 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1869 Epoch 310/500
1870 692/693 [============================>.] - ETA: 0s
      - loss: 0.0266 - accuracy: 0.9926
1871 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1872 693/693 [==============================] - 198s
     285ms/step - loss: 0.0267 - accuracy: 0.9926 -
     val_loss: 0.3080 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1873 Epoch 311/500
1874 692/693 [============================>.] - ETA: 0s
      - loss: 0.0245 - accuracy: 0.9930
1875 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1876 693/693 [==============================] - 198s
     285ms/step - loss: 0.0245 - accuracy: 0.9930 -
     val_loss: 0.3068 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1877 Epoch 312/500
1878 693/693 [==============================] - ETA: 0s
      - loss: 0.0286 - accuracy: 0.9917
1879 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1880 693/693 [==============================] - 198s
     285ms/step - loss: 0.0286 - accuracy: 0.9917 -
     val_loss: 0.3074 - val_accuracy: 0.9180 - lr: 1.
```

```
1880 0000e-05
1881 Epoch 313/500
1882 692/693 [============================>.] - ETA: 0s
      - loss: 0.0233 - accuracy: 0.9936
1883 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1884 693/693 [==============================] - 198s
     285ms/step - loss: 0.0236 - accuracy: 0.9935 -
     val_loss: 0.3068 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1885 Epoch 314/500
1886 692/693 [============================>.] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9929
1887 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1888 693/693 [==============================] - 198s
     285ms/step - loss: 0.0230 - accuracy: 0.9929 -
     val_loss: 0.3067 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1889 Epoch 315/500
1890 692/693 [============================>.] - ETA: 0s
      - loss: 0.0232 - accuracy: 0.9938
1891 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1892 693/693 [==============================] - 198s
     285ms/step - loss: 0.0232 - accuracy: 0.9938 -
     val_loss: 0.3064 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1893 Epoch 316/500
1894 692/693 [============================>.] - ETA: 0s
      - loss: 0.0225 - accuracy: 0.9935
1895 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1896 693/693 [==============================] - 198s
     285ms/step - loss: 0.0225 - accuracy: 0.9934 -
     val_loss: 0.3073 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1897 Epoch 317/500
1898 692/693 [============================>.] - ETA: 0s
      - loss: 0.0252 - accuracy: 0.9928
1899 [Plateau Check] Variação últimas 30 épocas: 0.
```

```
003975
693/693 [==============================] - 198s
285ms/step - loss: 0.0255 - accuracy: 0.9927 -
val_loss: 0.3072 - val_accuracy: 0.9198 - lr: 1.
0000e-05
Epoch 318/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0270 - accuracy: 0.9920
[Plateau Check] Variação últimas 30 épocas: 0.
003975
693/693 [==============================] - 198s
285ms/step - loss: 0.0270 - accuracy: 0.9920 -
val_loss: 0.3059 - val_accuracy: 0.9205 - lr: 1.
0000e-05
Epoch 319/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0243 - accuracy: 0.9931
[Plateau Check] Variação últimas 30 épocas: 0.
003253
693/693 [==============================] - 198s
285ms/step - loss: 0.0243 - accuracy: 0.9931 -
val_loss: 0.3072 - val_accuracy: 0.9176 - lr: 1.
0000e-05
Epoch 320/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0254 - accuracy: 0.9938
[Plateau Check] Variação últimas 30 épocas: 0.
003614
693/693 [==============================] - 198s
285ms/step - loss: 0.0254 - accuracy: 0.9938 -
val_loss: 0.3061 - val_accuracy: 0.9209 - lr: 1.
0000e-05
Epoch 321/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0223 - accuracy: 0.9949
[Plateau Check] Variação últimas 30 épocas: 0.
003614
693/693 [==============================] - 198s
285ms/step - loss: 0.0223 - accuracy: 0.9949 -
val_loss: 0.3057 - val_accuracy: 0.9190 - lr: 1.
0000e-05
```

```
1917 Epoch 322/500
1918 693/693 [==============================] - ETA: 0s
      - loss: 0.0256 - accuracy: 0.9920
1919 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1920 693/693 [==============================] - 198s
     285ms/step - loss: 0.0256 - accuracy: 0.9920 -
     val_loss: 0.3066 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1921 Epoch 323/500
1922 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0267 - accuracy: 0.9922
1923 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
1924 693/693 [==============================] - 198s
     285ms/step - loss: 0.0270 - accuracy: 0.9921 -
     val_loss: 0.3074 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1925 Epoch 324/500
1926 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0262 - accuracy: 0.9924
1927 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1928 693/693 [==============================] - 198s
     285ms/step - loss: 0.0263 - accuracy: 0.9923 -
     val_loss: 0.3073 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1929 Epoch 325/500
1930 693/693 [==============================] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9925
1931 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1932 693/693 [==============================] - 198s
     285ms/step - loss: 0.0249 - accuracy: 0.9925 -
     val_loss: 0.3060 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1933 Epoch 326/500
1934 693/693 [==============================] - ETA: 0s
      - loss: 0.0272 - accuracy: 0.9911
1935 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
```

```
1936 693/693 [==============================] - 198s
     285ms/step - loss: 0.0272 - accuracy: 0.9911 -
     val_loss: 0.3085 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1937 Epoch 327/500
1938 692/693 [============================>.] - ETA: 0s
      - loss: 0.0262 - accuracy: 0.9928
1939 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1940 693/693 [==============================] - 198s
     285ms/step - loss: 0.0262 - accuracy: 0.9928 -
     val_loss: 0.3072 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1941 Epoch 328/500
1942 693/693 [==============================] - ETA: 0s
      - loss: 0.0240 - accuracy: 0.9932
1943 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1944 693/693 [==============================] - 198s
     285ms/step - loss: 0.0240 - accuracy: 0.9932 -
     val_loss: 0.3067 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
1945 Epoch 329/500
1946 693/693 [==============================] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9929
1947 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1948 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9929 -
     val_loss: 0.3065 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1949 Epoch 330/500
1950 692/693 [============================>.] - ETA: 0s
      - loss: 0.0270 - accuracy: 0.9931
1951 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1952 693/693 [==============================] - 198s
     285ms/step - loss: 0.0270 - accuracy: 0.9931 -
     val_loss: 0.3077 - val_accuracy: 0.9172 - lr: 1.
     0000e-05
1953 Epoch 331/500
```

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1954 692/693 [============================>.] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9927
1955 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1956 693/693 [==============================] - 198s
     285ms/step - loss: 0.0244 - accuracy: 0.9927 -
     val_loss: 0.3073 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
1957 Epoch 332/500
1958 693/693 [==============================] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9937
1959 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1960 693/693 [==============================] - 198s
     285ms/step - loss: 0.0244 - accuracy: 0.9937 -
     val_loss: 0.3067 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1961 Epoch 333/500
1962 692/693 [============================>.] - ETA: 0s
      - loss: 0.0239 - accuracy: 0.9937
1963 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1964 693/693 [==============================] - 198s
     285ms/step - loss: 0.0240 - accuracy: 0.9936 -
     val_loss: 0.3079 - val_accuracy: 0.9212 - lr: 1.
     0000e-05
1965 Epoch 334/500
1966 693/693 [==============================] - ETA: 0s
      - loss: 0.0231 - accuracy: 0.9946
1967 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1968 693/693 [==============================] - 198s
     285ms/step - loss: 0.0231 - accuracy: 0.9946 -
     val_loss: 0.3077 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1969 Epoch 335/500
1970 692/693 [============================>.] - ETA: 0s
      - loss: 0.0238 - accuracy: 0.9924
1971 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1972 693/693 [==============================] - 198s
```

```
1972 285ms/step - loss: 0.0238 - accuracy: 0.9924 -
     val_loss: 0.3056 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
1973 Epoch 336/500
1974 692/693 [============================>.] - ETA: 0s
      - loss: 0.0238 - accuracy: 0.9942
1975 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1976 693/693 [==============================] - 198s
     285ms/step - loss: 0.0240 - accuracy: 0.9941 -
     val_loss: 0.3071 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1977 Epoch 337/500
1978 692/693 [============================>.] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9935
1979 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1980 693/693 [==============================] - 198s
     285ms/step - loss: 0.0233 - accuracy: 0.9934 -
     val_loss: 0.3058 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
1981 Epoch 338/500
1982 692/693 [============================>.] - ETA: 0s
      - loss: 0.0245 - accuracy: 0.9929
1983 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1984 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9928 -
     val_loss: 0.3079 - val_accuracy: 0.9176 - lr: 1.
     0000e-05
1985 Epoch 339/500
1986 692/693 [============================>.] - ETA: 0s
      - loss: 0.0239 - accuracy: 0.9932
1987 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1988 693/693 [==============================] - 198s
     285ms/step - loss: 0.0239 - accuracy: 0.9932 -
     val_loss: 0.3068 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
1989 Epoch 340/500
1990 693/693 [==============================] - ETA: 0s
```

```
1990  - loss: 0.0226 - accuracy: 0.9937
1991 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1992 693/693 [==============================] - 198s
     285ms/step - loss: 0.0226 - accuracy: 0.9937 -
     val_loss: 0.3059 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
1993 Epoch 341/500
1994 692/693 [============================>.] - ETA: 0s
      - loss: 0.0227 - accuracy: 0.9937
1995 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
1996 693/693 [==============================] - 198s
     285ms/step - loss: 0.0227 - accuracy: 0.9937 -
     val_loss: 0.3073 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
1997 Epoch 342/500
1998 692/693 [============================>.] - ETA: 0s
      - loss: 0.0225 - accuracy: 0.9942
1999 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2000 693/693 [==============================] - 199s
     285ms/step - loss: 0.0226 - accuracy: 0.9942 -
     val_loss: 0.3081 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2001 Epoch 343/500
2002 692/693 [============================>.] - ETA: 0s
      - loss: 0.0254 - accuracy: 0.9923
2003 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2004 693/693 [==============================] - 199s
     286ms/step - loss: 0.0254 - accuracy: 0.9923 -
     val_loss: 0.3078 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2005 Epoch 344/500
2006 692/693 [============================>.] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9936
2007 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2008 693/693 [==============================] - 198s
     285ms/step - loss: 0.0236 - accuracy: 0.9936 -
```

```
2008 val_loss: 0.3080 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2009 Epoch 345/500
2010 692/693 [============================>.] - ETA: 0s
      - loss: 0.0223 - accuracy: 0.9939
2011 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2012 693/693 [==============================] - 198s
     285ms/step - loss: 0.0239 - accuracy: 0.9938 -
     val_loss: 0.3083 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2013 Epoch 346/500
2014 693/693 [==============================] - ETA: 0s
      - loss: 0.0217 - accuracy: 0.9944
2015 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2016 693/693 [==============================] - 198s
     285ms/step - loss: 0.0217 - accuracy: 0.9944 -
     val_loss: 0.3066 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2017 Epoch 347/500
2018 692/693 [============================>.] - ETA: 0s
      - loss: 0.0234 - accuracy: 0.9939
2019 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2020 693/693 [==============================] - 198s
     285ms/step - loss: 0.0234 - accuracy: 0.9939 -
     val_loss: 0.3076 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2021 Epoch 348/500
2022 692/693 [============================>.] - ETA: 0s
      - loss: 0.0234 - accuracy: 0.9925
2023 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2024 693/693 [==============================] - 198s
     285ms/step - loss: 0.0234 - accuracy: 0.9925 -
     val_loss: 0.3085 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2025 Epoch 349/500
2026 692/693 [============================>.] - ETA: 0s
      - loss: 0.0245 - accuracy: 0.9935
```

```
2027 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2028 693/693 [==============================] - 198s
     285ms/step - loss: 0.0248 - accuracy: 0.9934 -
     val_loss: 0.3064 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2029 Epoch 350/500
2030 692/693 [============================>.] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9930
2031 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2032 693/693 [==============================] - 198s
     285ms/step - loss: 0.0237 - accuracy: 0.9930 -
     val_loss: 0.3076 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2033 Epoch 351/500
2034 692/693 [============================>.] - ETA: 0s
      - loss: 0.0227 - accuracy: 0.9939
2035 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2036 693/693 [==============================] - 198s
     285ms/step - loss: 0.0230 - accuracy: 0.9938 -
     val_loss: 0.3083 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2037 Epoch 352/500
2038 692/693 [============================>.] - ETA: 0s
      - loss: 0.0225 - accuracy: 0.9942
2039 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2040 693/693 [==============================] - 198s
     285ms/step - loss: 0.0225 - accuracy: 0.9942 -
     val_loss: 0.3070 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2041 Epoch 353/500
2042 692/693 [============================>.] - ETA: 0s
      - loss: 0.0227 - accuracy: 0.9943
2043 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
2044 693/693 [==============================] - 198s
     285ms/step - loss: 0.0229 - accuracy: 0.9942 -
     val_loss: 0.3060 - val_accuracy: 0.9219 - lr: 1.
```

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2044 0000e-05
2045 Epoch 354/500
2046 692/693 [============================>.] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9934
2047 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
2048 693/693 [==============================] - 198s
     285ms/step - loss: 0.0228 - accuracy: 0.9934 -
     val_loss: 0.3081 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2049 Epoch 355/500
2050 693/693 [==============================] - ETA: 0s
      - loss: 0.0241 - accuracy: 0.9935
2051 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
2052 693/693 [==============================] - 199s
     286ms/step - loss: 0.0241 - accuracy: 0.9935 -
     val_loss: 0.3078 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2053 Epoch 356/500
2054 692/693 [============================>.] - ETA: 0s
      - loss: 0.0229 - accuracy: 0.9939
2055 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
2056 693/693 [==============================] - 198s
     285ms/step - loss: 0.0235 - accuracy: 0.9937 -
     val_loss: 0.3096 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2057 Epoch 357/500
2058 693/693 [==============================] - ETA: 0s
      - loss: 0.0226 - accuracy: 0.9933
2059 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
2060 693/693 [==============================] - 198s
     285ms/step - loss: 0.0226 - accuracy: 0.9933 -
     val_loss: 0.3075 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2061 Epoch 358/500
2062 692/693 [============================>.] - ETA: 0s
      - loss: 0.0227 - accuracy: 0.9943
2063 [Plateau Check] Variação últimas 30 épocas: 0.
```

```
2063 004698
2064 693/693 [==============================] - 198s
     285ms/step - loss: 0.0228 - accuracy: 0.9942 -
     val_loss: 0.3084 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2065 Epoch 359/500
2066 692/693 [============================>.] - ETA: 0s
      - loss: 0.0256 - accuracy: 0.9926
2067 [Plateau Check] Variação últimas 30 épocas: 0.
     004698
2068 693/693 [==============================] - 198s
     285ms/step - loss: 0.0257 - accuracy: 0.9925 -
     val_loss: 0.3087 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2069 Epoch 360/500
2070 692/693 [============================>.] - ETA: 0s
      - loss: 0.0239 - accuracy: 0.9936
2071 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
2072 693/693 [==============================] - 198s
     285ms/step - loss: 0.0240 - accuracy: 0.9935 -
     val_loss: 0.3077 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2073 Epoch 361/500
2074 692/693 [============================>.] - ETA: 0s
      - loss: 0.0212 - accuracy: 0.9940
2075 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
2076 693/693 [==============================] - 198s
     285ms/step - loss: 0.0218 - accuracy: 0.9939 -
     val_loss: 0.3079 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2077 Epoch 362/500
2078 692/693 [============================>.] - ETA: 0s
      - loss: 0.0242 - accuracy: 0.9939
2079 [Plateau Check] Variação últimas 30 épocas: 0.
     004337
2080 693/693 [==============================] - 198s
     285ms/step - loss: 0.0242 - accuracy: 0.9939 -
     val_loss: 0.3078 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
```

```
Epoch 363/500
692/693 [============================>.] - ETA: 0s - loss: 0.0245 - accuracy: 0.9936
[Plateau Check] Variação últimas 30 épocas: 0.004337
693/693 [==============================] - 198s 285ms/step - loss: 0.0249 - accuracy: 0.9935 - val_loss: 0.3069 - val_accuracy: 0.9201 - lr: 1.0000e-05
Epoch 364/500
692/693 [============================>.] - ETA: 0s - loss: 0.0225 - accuracy: 0.9939
[Plateau Check] Variação últimas 30 épocas: 0.004337
693/693 [==============================] - 198s 285ms/step - loss: 0.0225 - accuracy: 0.9939 - val_loss: 0.3074 - val_accuracy: 0.9198 - lr: 1.0000e-05
Epoch 365/500
692/693 [============================>.] - ETA: 0s - loss: 0.0251 - accuracy: 0.9919
[Plateau Check] Variação últimas 30 épocas: 0.004337
693/693 [==============================] - 198s 285ms/step - loss: 0.0252 - accuracy: 0.9918 - val_loss: 0.3080 - val_accuracy: 0.9201 - lr: 1.0000e-05
Epoch 366/500
692/693 [============================>.] - ETA: 0s - loss: 0.0218 - accuracy: 0.9947
[Plateau Check] Variação últimas 30 épocas: 0.004337
693/693 [==============================] - 198s 285ms/step - loss: 0.0218 - accuracy: 0.9947 - val_loss: 0.3079 - val_accuracy: 0.9201 - lr: 1.0000e-05
Epoch 367/500
692/693 [============================>.] - ETA: 0s - loss: 0.0250 - accuracy: 0.9924
[Plateau Check] Variação últimas 30 épocas: 0.004337
```

```
2100 693/693 [==============================] - 198s
     285ms/step - loss: 0.0250 - accuracy: 0.9924 -
     val_loss: 0.3070 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2101 Epoch 368/500
2102 693/693 [==============================] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9939
2103 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2104 693/693 [==============================] - 198s
     285ms/step - loss: 0.0236 - accuracy: 0.9939 -
     val_loss: 0.3084 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2105 Epoch 369/500
2106 693/693 [==============================] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9933
2107 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2108 693/693 [==============================] - 198s
     285ms/step - loss: 0.0236 - accuracy: 0.9933 -
     val_loss: 0.3094 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
2109 Epoch 370/500
2110 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0240 - accuracy: 0.9935
2111 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2112 693/693 [==============================] - 198s
     285ms/step - loss: 0.0247 - accuracy: 0.9934 -
     val_loss: 0.3094 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2113 Epoch 371/500
2114 693/693 [==============================] - ETA: 0s
      - loss: 0.0278 - accuracy: 0.9919
2115 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2116 693/693 [==============================] - 198s
     285ms/step - loss: 0.0278 - accuracy: 0.9919 -
     val_loss: 0.3087 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2117 Epoch 372/500
```

```
2118 693/693 [==============================] - ETA: 0s
      - loss: 0.0234 - accuracy: 0.9935
2119 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2120 693/693 [==============================] - 199s
     286ms/step - loss: 0.0234 - accuracy: 0.9935 -
     val_loss: 0.3086 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
2121 Epoch 373/500
2122 693/693 [==============================] - ETA: 0s
      - loss: 0.0243 - accuracy: 0.9932
2123 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2124 693/693 [==============================] - 199s
     286ms/step - loss: 0.0243 - accuracy: 0.9932 -
     val_loss: 0.3076 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2125 Epoch 374/500
2126 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9940
2127 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2128 693/693 [==============================] - 199s
     286ms/step - loss: 0.0228 - accuracy: 0.9940 -
     val_loss: 0.3084 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
2129 Epoch 375/500
2130 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0255 - accuracy: 0.9930
2131 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2132 693/693 [==============================] - 199s
     285ms/step - loss: 0.0255 - accuracy: 0.9930 -
     val_loss: 0.3086 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2133 Epoch 376/500
2134 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9938
2135 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2136 693/693 [==============================] - 198s
```

```
2136 285ms/step - loss: 0.0238 - accuracy: 0.9937 -
     val_loss: 0.3091 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2137 Epoch 377/500
2138 692/693 [============================>.] - ETA: 0s
      - loss: 0.0247 - accuracy: 0.9928
2139 [Plateau Check] Variação últimas 30 épocas: 0.
     003975
2140 693/693 [==============================] - 198s
     285ms/step - loss: 0.0251 - accuracy: 0.9927 -
     val_loss: 0.3075 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
2141 Epoch 378/500
2142 692/693 [============================>.] - ETA: 0s
      - loss: 0.0223 - accuracy: 0.9934
2143 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2144 693/693 [==============================] - 198s
     285ms/step - loss: 0.0223 - accuracy: 0.9934 -
     val_loss: 0.3080 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2145 Epoch 379/500
2146 693/693 [==============================] - ETA: 0s
      - loss: 0.0254 - accuracy: 0.9930
2147 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2148 693/693 [==============================] - 199s
     285ms/step - loss: 0.0254 - accuracy: 0.9930 -
     val_loss: 0.3080 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2149 Epoch 380/500
2150 692/693 [============================>.] - ETA: 0s
      - loss: 0.0226 - accuracy: 0.9934
2151 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2152 693/693 [==============================] - 199s
     285ms/step - loss: 0.0227 - accuracy: 0.9933 -
     val_loss: 0.3087 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2153 Epoch 381/500
2154 692/693 [============================>.] - ETA: 0s
```

```
2154  - loss: 0.0223 - accuracy: 0.9936
2155 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2156 693/693 [==============================] - 198s
     285ms/step - loss: 0.0223 - accuracy: 0.9936 -
     val_loss: 0.3081 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2157 Epoch 382/500
2158 692/693 [============================>.] - ETA: 0s
      - loss: 0.0245 - accuracy: 0.9929
2159 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2160 693/693 [==============================] - 198s
     285ms/step - loss: 0.0245 - accuracy: 0.9929 -
     val_loss: 0.3078 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2161 Epoch 383/500
2162 692/693 [============================>.] - ETA: 0s
      - loss: 0.0248 - accuracy: 0.9935
2163 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2164 693/693 [==============================] - 198s
     285ms/step - loss: 0.0248 - accuracy: 0.9935 -
     val_loss: 0.3092 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2165 Epoch 384/500
2166 693/693 [==============================] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9935
2167 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2168 693/693 [==============================] - 198s
     285ms/step - loss: 0.0244 - accuracy: 0.9935 -
     val_loss: 0.3079 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2169 Epoch 385/500
2170 692/693 [============================>.] - ETA: 0s
      - loss: 0.0254 - accuracy: 0.9925
2171 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2172 693/693 [==============================] - 199s
     286ms/step - loss: 0.0258 - accuracy: 0.9924 -
```

```
2172 val_loss: 0.3087 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2173 Epoch 386/500
2174 692/693 [============================>.] - ETA: 0s
      - loss: 0.0234 - accuracy: 0.9930
2175 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2176 693/693 [==============================] - 199s
     286ms/step - loss: 0.0235 - accuracy: 0.9930 -
     val_loss: 0.3096 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2177 Epoch 387/500
2178 692/693 [============================>.] - ETA: 0s
      - loss: 0.0247 - accuracy: 0.9934
2179 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2180 693/693 [==============================] - 199s
     286ms/step - loss: 0.0248 - accuracy: 0.9934 -
     val_loss: 0.3083 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2181 Epoch 388/500
2182 692/693 [============================>.] - ETA: 0s
      - loss: 0.0226 - accuracy: 0.9935
2183 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2184 693/693 [==============================] - 199s
     285ms/step - loss: 0.0226 - accuracy: 0.9935 -
     val_loss: 0.3081 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2185 Epoch 389/500
2186 692/693 [============================>.] - ETA: 0s
      - loss: 0.0242 - accuracy: 0.9929
2187 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2188 693/693 [==============================] - 198s
     285ms/step - loss: 0.0242 - accuracy: 0.9929 -
     val_loss: 0.3085 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2189 Epoch 390/500
2190 692/693 [============================>.] - ETA: 0s
      - loss: 0.0203 - accuracy: 0.9943
```

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2191 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2192 693/693 [==============================] - 198s
     285ms/step - loss: 0.0203 - accuracy: 0.9943 -
     val_loss: 0.3077 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2193 Epoch 391/500
2194 693/693 [==============================] - ETA: 0s
      - loss: 0.0256 - accuracy: 0.9931
2195 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2196 693/693 [==============================] - 199s
     286ms/step - loss: 0.0256 - accuracy: 0.9931 -
     val_loss: 0.3076 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2197 Epoch 392/500
2198 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0254 - accuracy: 0.9924
2199 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2200 693/693 [==============================] - 199s
     286ms/step - loss: 0.0257 - accuracy: 0.9923 -
     val_loss: 0.3074 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2201 Epoch 393/500
2202 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0254 - accuracy: 0.9929
2203 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2204 693/693 [==============================] - 199s
     286ms/step - loss: 0.0254 - accuracy: 0.9928 -
     val_loss: 0.3077 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2205 Epoch 394/500
2206 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0220 - accuracy: 0.9949
2207 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2208 693/693 [==============================] - 199s
     286ms/step - loss: 0.0223 - accuracy: 0.9948 -
     val_loss: 0.3093 - val_accuracy: 0.9198 - lr: 1.
```

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0000e-05
Epoch 395/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0238 - accuracy: 0.9938
[Plateau Check] Variação últimas 30 épocas: 0.
002168
693/693 [==============================] - 198s
285ms/step - loss: 0.0243 - accuracy: 0.9937 -
val_loss: 0.3080 - val_accuracy: 0.9198 - lr: 1.
0000e-05
Epoch 396/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0247 - accuracy: 0.9931
[Plateau Check] Variação últimas 30 épocas: 0.
002168
693/693 [==============================] - 199s
285ms/step - loss: 0.0247 - accuracy: 0.9931 -
val_loss: 0.3083 - val_accuracy: 0.9194 - lr: 1.
0000e-05
Epoch 397/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0240 - accuracy: 0.9933
[Plateau Check] Variação últimas 30 épocas: 0.
002891
693/693 [==============================] - 198s
285ms/step - loss: 0.0240 - accuracy: 0.9933 -
val_loss: 0.3096 - val_accuracy: 0.9176 - lr: 1.
0000e-05
Epoch 398/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0238 - accuracy: 0.9937
[Plateau Check] Variação últimas 30 épocas: 0.
002891
693/693 [==============================] - 199s
286ms/step - loss: 0.0238 - accuracy: 0.9937 -
val_loss: 0.3094 - val_accuracy: 0.9183 - lr: 1.
0000e-05
Epoch 399/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0243 - accuracy: 0.9935
[Plateau Check] Variação últimas 30 épocas: 0.
```

```
002891
693/693 [==============================] - 199s
    286ms/step - loss: 0.0243 - accuracy: 0.9935 -
    val_loss: 0.3100 - val_accuracy: 0.9190 - lr: 1.
    0000e-05
Epoch 400/500
693/693 [==============================] - ETA: 0s
     - loss: 0.0252 - accuracy: 0.9940
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 199s
    286ms/step - loss: 0.0252 - accuracy: 0.9940 -
    val_loss: 0.3101 - val_accuracy: 0.9198 - lr: 1.
    0000e-05
Epoch 401/500
692/693 [==============================>.] - ETA: 0s
     - loss: 0.0207 - accuracy: 0.9949
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 198s
    285ms/step - loss: 0.0210 - accuracy: 0.9948 -
    val_loss: 0.3090 - val_accuracy: 0.9190 - lr: 1.
    0000e-05
Epoch 402/500
692/693 [==============================>.] - ETA: 0s
     - loss: 0.0262 - accuracy: 0.9920
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 198s
    285ms/step - loss: 0.0262 - accuracy: 0.9920 -
    val_loss: 0.3087 - val_accuracy: 0.9180 - lr: 1.
    0000e-05
Epoch 403/500
692/693 [==============================>.] - ETA: 0s
     - loss: 0.0230 - accuracy: 0.9940
[Plateau Check] Variação últimas 30 épocas: 0.
    002891
693/693 [==============================] - 199s
    286ms/step - loss: 0.0231 - accuracy: 0.9940 -
    val_loss: 0.3098 - val_accuracy: 0.9194 - lr: 1.
    0000e-05
```

```
Epoch 404/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0247 - accuracy: 0.9931
[Plateau Check] Variação últimas 30 épocas: 0.
002530
693/693 [==============================] - 199s
286ms/step - loss: 0.0247 - accuracy: 0.9931 -
val_loss: 0.3082 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 405/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0233 - accuracy: 0.9934
[Plateau Check] Variação últimas 30 épocas: 0.
002530
693/693 [==============================] - 199s
286ms/step - loss: 0.0237 - accuracy: 0.9933 -
val_loss: 0.3084 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 406/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0217 - accuracy: 0.9938
[Plateau Check] Variação últimas 30 épocas: 0.
002530
693/693 [==============================] - 200s
287ms/step - loss: 0.0217 - accuracy: 0.9938 -
val_loss: 0.3097 - val_accuracy: 0.9190 - lr: 1.
0000e-05
Epoch 407/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0242 - accuracy: 0.9925
[Plateau Check] Variação últimas 30 épocas: 0.
002168
693/693 [==============================] - 199s
286ms/step - loss: 0.0242 - accuracy: 0.9925 -
val_loss: 0.3084 - val_accuracy: 0.9187 - lr: 1.
0000e-05
Epoch 408/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0235 - accuracy: 0.9934
[Plateau Check] Variação últimas 30 épocas: 0.
002168
```

```
2264 693/693 [==============================] - 199s
     286ms/step - loss: 0.0235 - accuracy: 0.9934 -
     val_loss: 0.3092 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2265 Epoch 409/500
2266 693/693 [==============================] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9929
2267 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2268 693/693 [==============================] - 199s
     285ms/step - loss: 0.0237 - accuracy: 0.9929 -
     val_loss: 0.3090 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2269 Epoch 410/500
2270 693/693 [==============================] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9934
2271 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2272 693/693 [==============================] - 199s
     286ms/step - loss: 0.0230 - accuracy: 0.9934 -
     val_loss: 0.3085 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2273 Epoch 411/500
2274 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9928
2275 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2276 693/693 [==============================] - 199s
     286ms/step - loss: 0.0230 - accuracy: 0.9927 -
     val_loss: 0.3092 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2277 Epoch 412/500
2278 693/693 [==============================] - ETA: 0s
      - loss: 0.0238 - accuracy: 0.9930
2279 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2280 693/693 [==============================] - 199s
     285ms/step - loss: 0.0238 - accuracy: 0.9930 -
     val_loss: 0.3097 - val_accuracy: 0.9169 - lr: 1.
     0000e-05
2281 Epoch 413/500
```

```
2282 692/693 [============================>.] - ETA: 0s
      - loss: 0.0235 - accuracy: 0.9932
2283 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2284 693/693 [==============================] - 199s
     286ms/step - loss: 0.0237 - accuracy: 0.9931 -
     val_loss: 0.3095 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2285 Epoch 414/500
2286 692/693 [============================>.] - ETA: 0s
      - loss: 0.0221 - accuracy: 0.9946
2287 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2288 693/693 [==============================] - 199s
     286ms/step - loss: 0.0221 - accuracy: 0.9946 -
     val_loss: 0.3092 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2289 Epoch 415/500
2290 693/693 [==============================] - ETA: 0s
      - loss: 0.0237 - accuracy: 0.9934
2291 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2292 693/693 [==============================] - 199s
     286ms/step - loss: 0.0237 - accuracy: 0.9934 -
     val_loss: 0.3094 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2293 Epoch 416/500
2294 693/693 [==============================] - ETA: 0s
      - loss: 0.0227 - accuracy: 0.9936
2295 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2296 693/693 [==============================] - 199s
     286ms/step - loss: 0.0227 - accuracy: 0.9936 -
     val_loss: 0.3080 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2297 Epoch 417/500
2298 693/693 [==============================] - ETA: 0s
      - loss: 0.0222 - accuracy: 0.9941
2299 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2300 693/693 [==============================] - 199s
```

```
2300 286ms/step - loss: 0.0222 - accuracy: 0.9941 -
     val_loss: 0.3087 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2301 Epoch 418/500
2302 693/693 [==============================] - ETA: 0s
      - loss: 0.0258 - accuracy: 0.9921
2303 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2304 693/693 [==============================] - 199s
     286ms/step - loss: 0.0258 - accuracy: 0.9921 -
     val_loss: 0.3100 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2305 Epoch 419/500
2306 693/693 [==============================] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9930
2307 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2308 693/693 [==============================] - 199s
     286ms/step - loss: 0.0244 - accuracy: 0.9930 -
     val_loss: 0.3083 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2309 Epoch 420/500
2310 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0224 - accuracy: 0.9944
2311 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2312 693/693 [==============================] - 199s
     286ms/step - loss: 0.0239 - accuracy: 0.9942 -
     val_loss: 0.3104 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2313 Epoch 421/500
2314 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0209 - accuracy: 0.9948
2315 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2316 693/693 [==============================] - 199s
     286ms/step - loss: 0.0209 - accuracy: 0.9948 -
     val_loss: 0.3113 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2317 Epoch 422/500
2318 693/693 [==============================] - ETA: 0s
```

```
2318  - loss: 0.0222 - accuracy: 0.9937
2319 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2320 693/693 [==============================] - 199s
     286ms/step - loss: 0.0222 - accuracy: 0.9937 -
     val_loss: 0.3095 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2321 Epoch 423/500
2322 692/693 [============================>.] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9930
2323 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2324 693/693 [==============================] - 199s
     286ms/step - loss: 0.0249 - accuracy: 0.9930 -
     val_loss: 0.3087 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2325 Epoch 424/500
2326 693/693 [==============================] - ETA: 0s
      - loss: 0.0219 - accuracy: 0.9939
2327 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2328 693/693 [==============================] - 199s
     285ms/step - loss: 0.0219 - accuracy: 0.9939 -
     val_loss: 0.3085 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2329 Epoch 425/500
2330 693/693 [==============================] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9936
2331 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2332 693/693 [==============================] - 199s
     285ms/step - loss: 0.0250 - accuracy: 0.9936 -
     val_loss: 0.3102 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2333 Epoch 426/500
2334 692/693 [============================>.] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9942
2335 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2336 693/693 [==============================] - 199s
     285ms/step - loss: 0.0230 - accuracy: 0.9942 -
```

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2336 val_loss: 0.3090 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2337 Epoch 427/500
2338 692/693 [============================>.] - ETA: 0s
      - loss: 0.0222 - accuracy: 0.9939
2339 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2340 693/693 [==============================] - 199s
     285ms/step - loss: 0.0222 - accuracy: 0.9939 -
     val_loss: 0.3099 - val_accuracy: 0.9205 - lr: 1.
     0000e-05
2341 Epoch 428/500
2342 692/693 [============================>.] - ETA: 0s
      - loss: 0.0225 - accuracy: 0.9943
2343 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2344 693/693 [==============================] - 199s
     285ms/step - loss: 0.0225 - accuracy: 0.9943 -
     val_loss: 0.3088 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2345 Epoch 429/500
2346 693/693 [==============================] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9927
2347 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2348 693/693 [==============================] - 199s
     286ms/step - loss: 0.0230 - accuracy: 0.9927 -
     val_loss: 0.3095 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2349 Epoch 430/500
2350 692/693 [============================>.] - ETA: 0s
      - loss: 0.0223 - accuracy: 0.9939
2351 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2352 693/693 [==============================] - 199s
     286ms/step - loss: 0.0223 - accuracy: 0.9939 -
     val_loss: 0.3071 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2353 Epoch 431/500
2354 692/693 [============================>.] - ETA: 0s
      - loss: 0.0247 - accuracy: 0.9929
```

```
2355 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2356 693/693 [==============================] - 199s
     286ms/step - loss: 0.0252 - accuracy: 0.9928 -
     val_loss: 0.3095 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2357 Epoch 432/500
2358 692/693 [============================>.] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9946
2359 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2360 693/693 [==============================] - 199s
     286ms/step - loss: 0.0230 - accuracy: 0.9944 -
     val_loss: 0.3101 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2361 Epoch 433/500
2362 692/693 [============================>.] - ETA: 0s
      - loss: 0.0243 - accuracy: 0.9934
2363 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2364 693/693 [==============================] - 199s
     286ms/step - loss: 0.0243 - accuracy: 0.9934 -
     val_loss: 0.3101 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2365 Epoch 434/500
2366 692/693 [============================>.] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9928
2367 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2368 693/693 [==============================] - 199s
     286ms/step - loss: 0.0257 - accuracy: 0.9927 -
     val_loss: 0.3088 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2369 Epoch 435/500
2370 693/693 [==============================] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9937
2371 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2372 693/693 [==============================] - 199s
     285ms/step - loss: 0.0236 - accuracy: 0.9937 -
     val_loss: 0.3102 - val_accuracy: 0.9198 - lr: 1.
```

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2372 0000e-05
2373 Epoch 436/500
2374 692/693 [============================>.] - ETA: 0s
      - loss: 0.0248 - accuracy: 0.9931
2375 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2376 693/693 [==============================] - 199s
     285ms/step - loss: 0.0248 - accuracy: 0.9931 -
     val_loss: 0.3085 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2377 Epoch 437/500
2378 692/693 [============================>.] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9930
2379 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2380 693/693 [==============================] - 199s
     287ms/step - loss: 0.0249 - accuracy: 0.9930 -
     val_loss: 0.3094 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2381 Epoch 438/500
2382 693/693 [==============================] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9926
2383 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2384 693/693 [==============================] - 199s
     286ms/step - loss: 0.0244 - accuracy: 0.9926 -
     val_loss: 0.3103 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2385 Epoch 439/500
2386 692/693 [============================>.] - ETA: 0s
      - loss: 0.0226 - accuracy: 0.9935
2387 [Plateau Check] Variação últimas 30 épocas: 0.
     003614
2388 693/693 [==============================] - 199s
     285ms/step - loss: 0.0226 - accuracy: 0.9935 -
     val_loss: 0.3111 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2389 Epoch 440/500
2390 693/693 [==============================] - ETA: 0s
      - loss: 0.0223 - accuracy: 0.9943
2391 [Plateau Check] Variação últimas 30 épocas: 0.
```

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003614
693/693 [==============================] - 199s
    286ms/step - loss: 0.0223 - accuracy: 0.9943 -
    val_loss: 0.3100 - val_accuracy: 0.9201 - lr: 1.
    0000e-05
Epoch 441/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0233 - accuracy: 0.9930
[Plateau Check] Variação últimas 30 épocas: 0.
    003614
693/693 [==============================] - 199s
    286ms/step - loss: 0.0233 - accuracy: 0.9930 -
    val_loss: 0.3095 - val_accuracy: 0.9198 - lr: 1.
    0000e-05
Epoch 442/500
693/693 [==============================] - ETA: 0s
     - loss: 0.0220 - accuracy: 0.9935
[Plateau Check] Variação últimas 30 épocas: 0.
    002530
693/693 [==============================] - 199s
    285ms/step - loss: 0.0220 - accuracy: 0.9935 -
    val_loss: 0.3092 - val_accuracy: 0.9201 - lr: 1.
    0000e-05
Epoch 443/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0253 - accuracy: 0.9928
[Plateau Check] Variação últimas 30 épocas: 0.
    002530
693/693 [==============================] - 199s
    285ms/step - loss: 0.0254 - accuracy: 0.9928 -
    val_loss: 0.3091 - val_accuracy: 0.9187 - lr: 1.
    0000e-05
Epoch 444/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0228 - accuracy: 0.9936
[Plateau Check] Variação últimas 30 épocas: 0.
    002530
693/693 [==============================] - 199s
    286ms/step - loss: 0.0228 - accuracy: 0.9936 -
    val_loss: 0.3095 - val_accuracy: 0.9190 - lr: 1.
    0000e-05
```

```
2409 Epoch 445/500
2410 692/693 [============================>.] - ETA: 0s
      - loss: 0.0250 - accuracy: 0.9930
2411 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2412 693/693 [==============================] - 199s
     285ms/step - loss: 0.0266 - accuracy: 0.9929 -
     val_loss: 0.3100 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
2413 Epoch 446/500
2414 693/693 [==============================] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9940
2415 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2416 693/693 [==============================] - 199s
     285ms/step - loss: 0.0228 - accuracy: 0.9940 -
     val_loss: 0.3101 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2417 Epoch 447/500
2418 692/693 [============================>.] - ETA: 0s
      - loss: 0.0201 - accuracy: 0.9951
2419 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2420 693/693 [==============================] - 199s
     285ms/step - loss: 0.0201 - accuracy: 0.9951 -
     val_loss: 0.3111 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2421 Epoch 448/500
2422 693/693 [==============================] - ETA: 0s
      - loss: 0.0227 - accuracy: 0.9945
2423 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2424 693/693 [==============================] - 199s
     286ms/step - loss: 0.0227 - accuracy: 0.9945 -
     val_loss: 0.3103 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2425 Epoch 449/500
2426 693/693 [==============================] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9941
2427 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
```

```
2428 693/693 [==============================] - 199s
     286ms/step - loss: 0.0230 - accuracy: 0.9941 -
     val_loss: 0.3108 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2429 Epoch 450/500
2430 693/693 [==============================] - ETA: 0s
      - loss: 0.0219 - accuracy: 0.9941
2431 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2432 693/693 [==============================] - 199s
     286ms/step - loss: 0.0219 - accuracy: 0.9941 -
     val_loss: 0.3101 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2433 Epoch 451/500
2434 693/693 [==============================] - ETA: 0s
      - loss: 0.0222 - accuracy: 0.9939
2435 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2436 693/693 [==============================] - 198s
     285ms/step - loss: 0.0222 - accuracy: 0.9939 -
     val_loss: 0.3084 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2437 Epoch 452/500
2438 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0280 - accuracy: 0.9923
2439 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2440 693/693 [==============================] - 199s
     286ms/step - loss: 0.0280 - accuracy: 0.9923 -
     val_loss: 0.3102 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2441 Epoch 453/500
2442 693/693 [==============================] - ETA: 0s
      - loss: 0.0220 - accuracy: 0.9939
2443 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2444 693/693 [==============================] - 199s
     285ms/step - loss: 0.0220 - accuracy: 0.9939 -
     val_loss: 0.3093 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2445 Epoch 454/500
```

```
2446 693/693 [==============================] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9939
2447 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2448 693/693 [==============================] - 199s
     286ms/step - loss: 0.0228 - accuracy: 0.9939 -
     val_loss: 0.3110 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2449 Epoch 455/500
2450 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9942
2451 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2452 693/693 [==============================] - 199s
     285ms/step - loss: 0.0253 - accuracy: 0.9940 -
     val_loss: 0.3109 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2453 Epoch 456/500
2454 693/693 [==============================] - ETA: 0s
      - loss: 0.0205 - accuracy: 0.9945
2455 [Plateau Check] Variação últimas 30 épocas: 0.
     002530
2456 693/693 [==============================] - 199s
     286ms/step - loss: 0.0205 - accuracy: 0.9945 -
     val_loss: 0.3089 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2457 Epoch 457/500
2458 692/693 [==============================>.] - ETA: 0s
      - loss: 0.0231 - accuracy: 0.9932
2459 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2460 693/693 [==============================] - 199s
     285ms/step - loss: 0.0231 - accuracy: 0.9932 -
     val_loss: 0.3088 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2461 Epoch 458/500
2462 693/693 [==============================] - ETA: 0s
      - loss: 0.0245 - accuracy: 0.9930
2463 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2464 693/693 [==============================] - 199s
```

```
2464 286ms/step - loss: 0.0245 - accuracy: 0.9930 -
     val_loss: 0.3118 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
2465 Epoch 459/500
2466 692/693 [============================>.] - ETA: 0s
      - loss: 0.0255 - accuracy: 0.9928
2467 [Plateau Check] Variação últimas 30 épocas: 0.
     002168
2468 693/693 [==============================] - 199s
     286ms/step - loss: 0.0255 - accuracy: 0.9928 -
     val_loss: 0.3101 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
2469 Epoch 460/500
2470 693/693 [==============================] - ETA: 0s
      - loss: 0.0201 - accuracy: 0.9952
2471 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2472 693/693 [==============================] - 199s
     286ms/step - loss: 0.0201 - accuracy: 0.9952 -
     val_loss: 0.3108 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2473 Epoch 461/500
2474 693/693 [==============================] - ETA: 0s
      - loss: 0.0225 - accuracy: 0.9931
2475 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2476 693/693 [==============================] - 199s
     286ms/step - loss: 0.0225 - accuracy: 0.9931 -
     val_loss: 0.3096 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2477 Epoch 462/500
2478 693/693 [==============================] - ETA: 0s
      - loss: 0.0235 - accuracy: 0.9931
2479 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2480 693/693 [==============================] - 199s
     286ms/step - loss: 0.0235 - accuracy: 0.9931 -
     val_loss: 0.3104 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2481 Epoch 463/500
2482 692/693 [============================>.] - ETA: 0s
```

```
2482  - loss: 0.0231 - accuracy: 0.9934
2483 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2484 693/693 [==============================] - 199s
     286ms/step - loss: 0.0231 - accuracy: 0.9934 -
     val_loss: 0.3110 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2485 Epoch 464/500
2486 692/693 [============================>.] - ETA: 0s
      - loss: 0.0244 - accuracy: 0.9930
2487 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2488 693/693 [==============================] - 199s
     286ms/step - loss: 0.0248 - accuracy: 0.9929 -
     val_loss: 0.3097 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2489 Epoch 465/500
2490 692/693 [============================>.] - ETA: 0s
      - loss: 0.0236 - accuracy: 0.9930
2491 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2492 693/693 [==============================] - 199s
     286ms/step - loss: 0.0236 - accuracy: 0.9930 -
     val_loss: 0.3115 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2493 Epoch 466/500
2494 692/693 [============================>.] - ETA: 0s
      - loss: 0.0223 - accuracy: 0.9938
2495 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2496 693/693 [==============================] - 199s
     286ms/step - loss: 0.0224 - accuracy: 0.9937 -
     val_loss: 0.3109 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2497 Epoch 467/500
2498 692/693 [============================>.] - ETA: 0s
      - loss: 0.0248 - accuracy: 0.9933
2499 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2500 693/693 [==============================] - 199s
     286ms/step - loss: 0.0248 - accuracy: 0.9933 -
```

```
2500 val_loss: 0.3084 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2501 Epoch 468/500
2502 692/693 [============================>.] - ETA: 0s
      - loss: 0.0228 - accuracy: 0.9939
2503 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2504 693/693 [==============================] - 199s
     286ms/step - loss: 0.0233 - accuracy: 0.9938 -
     val_loss: 0.3108 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2505 Epoch 469/500
2506 692/693 [============================>.] - ETA: 0s
      - loss: 0.0217 - accuracy: 0.9941
2507 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2508 693/693 [==============================] - 199s
     286ms/step - loss: 0.0217 - accuracy: 0.9941 -
     val_loss: 0.3103 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2509 Epoch 470/500
2510 693/693 [==============================] - ETA: 0s
      - loss: 0.0222 - accuracy: 0.9949
2511 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2512 693/693 [==============================] - 199s
     286ms/step - loss: 0.0222 - accuracy: 0.9949 -
     val_loss: 0.3110 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2513 Epoch 471/500
2514 693/693 [==============================] - ETA: 0s
      - loss: 0.0220 - accuracy: 0.9942
2515 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2516 693/693 [==============================] - 199s
     286ms/step - loss: 0.0220 - accuracy: 0.9942 -
     val_loss: 0.3089 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2517 Epoch 472/500
2518 692/693 [============================>.] - ETA: 0s
      - loss: 0.0233 - accuracy: 0.9930
```

```
2519 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2520 693/693 [==============================] - 199s
     285ms/step - loss: 0.0233 - accuracy: 0.9930 -
     val_loss: 0.3100 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2521 Epoch 473/500
2522 692/693 [============================>.] - ETA: 0s
      - loss: 0.0229 - accuracy: 0.9939
2523 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2524 693/693 [==============================] - 199s
     285ms/step - loss: 0.0232 - accuracy: 0.9938 -
     val_loss: 0.3112 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2525 Epoch 474/500
2526 693/693 [==============================] - ETA: 0s
      - loss: 0.0253 - accuracy: 0.9934
2527 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2528 693/693 [==============================] - 199s
     286ms/step - loss: 0.0253 - accuracy: 0.9934 -
     val_loss: 0.3101 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2529 Epoch 475/500
2530 693/693 [==============================] - ETA: 0s
      - loss: 0.0226 - accuracy: 0.9931
2531 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2532 693/693 [==============================] - 199s
     286ms/step - loss: 0.0226 - accuracy: 0.9931 -
     val_loss: 0.3110 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2533 Epoch 476/500
2534 693/693 [==============================] - ETA: 0s
      - loss: 0.0210 - accuracy: 0.9941
2535 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2536 693/693 [==============================] - 199s
     286ms/step - loss: 0.0210 - accuracy: 0.9941 -
     val_loss: 0.3100 - val_accuracy: 0.9187 - lr: 1.
```

```
2536 0000e-05
2537 Epoch 477/500
2538 692/693 [============================>.] - ETA: 0s
      - loss: 0.0260 - accuracy: 0.9921
2539 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2540 693/693 [==============================] - 199s
     286ms/step - loss: 0.0266 - accuracy: 0.9921 -
     val_loss: 0.3101 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2541 Epoch 478/500
2542 692/693 [============================>.] - ETA: 0s
      - loss: 0.0235 - accuracy: 0.9931
2543 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2544 693/693 [==============================] - 199s
     286ms/step - loss: 0.0239 - accuracy: 0.9930 -
     val_loss: 0.3107 - val_accuracy: 0.9183 - lr: 1.
     0000e-05
2545 Epoch 479/500
2546 692/693 [============================>.] - ETA: 0s
      - loss: 0.0239 - accuracy: 0.9934
2547 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2548 693/693 [==============================] - 199s
     285ms/step - loss: 0.0246 - accuracy: 0.9933 -
     val_loss: 0.3101 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2549 Epoch 480/500
2550 692/693 [============================>.] - ETA: 0s
      - loss: 0.0208 - accuracy: 0.9946
2551 [Plateau Check] Variação últimas 30 épocas: 0.
     001807
2552 693/693 [==============================] - 199s
     286ms/step - loss: 0.0209 - accuracy: 0.9945 -
     val_loss: 0.3115 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2553 Epoch 481/500
2554 692/693 [============================>.] - ETA: 0s
      - loss: 0.0223 - accuracy: 0.9934
2555 [Plateau Check] Variação últimas 30 épocas: 0.
```

```
001807
693/693 [==============================] - 199s
    286ms/step - loss: 0.0228 - accuracy: 0.9933 -
    val_loss: 0.3113 - val_accuracy: 0.9183 - lr: 1.
    0000e-05
Epoch 482/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0217 - accuracy: 0.9939
[Plateau Check] Variação últimas 30 épocas: 0.
    001807
693/693 [==============================] - 199s
    286ms/step - loss: 0.0218 - accuracy: 0.9939 -
    val_loss: 0.3093 - val_accuracy: 0.9187 - lr: 1.
    0000e-05
Epoch 483/500
693/693 [==============================] - ETA: 0s
     - loss: 0.0231 - accuracy: 0.9941
[Plateau Check] Variação últimas 30 épocas: 0.
    002168
693/693 [==============================] - 199s
    286ms/step - loss: 0.0231 - accuracy: 0.9941 -
    val_loss: 0.3102 - val_accuracy: 0.9180 - lr: 1.
    0000e-05
Epoch 484/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0238 - accuracy: 0.9931
[Plateau Check] Variação últimas 30 épocas: 0.
    002168
693/693 [==============================] - 199s
    286ms/step - loss: 0.0239 - accuracy: 0.9931 -
    val_loss: 0.3107 - val_accuracy: 0.9194 - lr: 1.
    0000e-05
Epoch 485/500
692/693 [============================>.] - ETA: 0s
     - loss: 0.0224 - accuracy: 0.9930
[Plateau Check] Variação últimas 30 épocas: 0.
    002168
693/693 [==============================] - 199s
    286ms/step - loss: 0.0240 - accuracy: 0.9929 -
    val_loss: 0.3126 - val_accuracy: 0.9187 - lr: 1.
    0000e-05
```

```
Epoch 486/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0214 - accuracy: 0.9946
[Plateau Check] Variação últimas 30 épocas: 0.
002168
693/693 [==============================] - 199s
286ms/step - loss: 0.0217 - accuracy: 0.9945 -
val_loss: 0.3111 - val_accuracy: 0.9201 - lr: 1.
0000e-05
Epoch 487/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0234 - accuracy: 0.9940
[Plateau Check] Variação últimas 30 épocas: 0.
002168
693/693 [==============================] - 199s
286ms/step - loss: 0.0234 - accuracy: 0.9940 -
val_loss: 0.3117 - val_accuracy: 0.9194 - lr: 1.
0000e-05
Epoch 488/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0234 - accuracy: 0.9937
[Plateau Check] Variação últimas 30 épocas: 0.
002168
693/693 [==============================] - 199s
286ms/step - loss: 0.0236 - accuracy: 0.9936 -
val_loss: 0.3101 - val_accuracy: 0.9198 - lr: 1.
0000e-05
Epoch 489/500
692/693 [============================>.] - ETA: 0s
 - loss: 0.0217 - accuracy: 0.9939
[Plateau Check] Variação últimas 30 épocas: 0.
002891
693/693 [==============================] - 199s
286ms/step - loss: 0.0219 - accuracy: 0.9939 -
val_loss: 0.3098 - val_accuracy: 0.9209 - lr: 1.
0000e-05
Epoch 490/500
693/693 [==============================] - ETA: 0s
 - loss: 0.0233 - accuracy: 0.9937
[Plateau Check] Variação últimas 30 épocas: 0.
002891
```

```
2592 693/693 [==============================] - 199s
     286ms/step - loss: 0.0233 - accuracy: 0.9937 -
     val_loss: 0.3114 - val_accuracy: 0.9187 - lr: 1.
     0000e-05
2593 Epoch 491/500
2594 692/693 [============================>.] - ETA: 0s
      - loss: 0.0249 - accuracy: 0.9927
2595 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2596 693/693 [==============================] - 199s
     285ms/step - loss: 0.0249 - accuracy: 0.9927 -
     val_loss: 0.3104 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2597 Epoch 492/500
2598 692/693 [============================>.] - ETA: 0s
      - loss: 0.0218 - accuracy: 0.9944
2599 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2600 693/693 [==============================] - 199s
     286ms/step - loss: 0.0221 - accuracy: 0.9943 -
     val_loss: 0.3091 - val_accuracy: 0.9198 - lr: 1.
     0000e-05
2601 Epoch 493/500
2602 693/693 [==============================] - ETA: 0s
      - loss: 0.0234 - accuracy: 0.9929
2603 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2604 693/693 [==============================] - 199s
     286ms/step - loss: 0.0234 - accuracy: 0.9929 -
     val_loss: 0.3104 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2605 Epoch 494/500
2606 693/693 [==============================] - ETA: 0s
      - loss: 0.0233 - accuracy: 0.9936
2607 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2608 693/693 [==============================] - 199s
     286ms/step - loss: 0.0233 - accuracy: 0.9936 -
     val_loss: 0.3103 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2609 Epoch 495/500
```

```
2610 692/693 [============================>.] - ETA: 0s
      - loss: 0.0248 - accuracy: 0.9920
2611 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2612 693/693 [==============================] - 199s
     286ms/step - loss: 0.0248 - accuracy: 0.9920 -
     val_loss: 0.3101 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2613 Epoch 496/500
2614 692/693 [============================>.] - ETA: 0s
      - loss: 0.0230 - accuracy: 0.9942
2615 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2616 693/693 [==============================] - 199s
     286ms/step - loss: 0.0230 - accuracy: 0.9942 -
     val_loss: 0.3097 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2617 Epoch 497/500
2618 693/693 [==============================] - ETA: 0s
      - loss: 0.0217 - accuracy: 0.9940
2619 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2620 693/693 [==============================] - 199s
     286ms/step - loss: 0.0217 - accuracy: 0.9940 -
     val_loss: 0.3112 - val_accuracy: 0.9190 - lr: 1.
     0000e-05
2621 Epoch 498/500
2622 692/693 [============================>.] - ETA: 0s
      - loss: 0.0231 - accuracy: 0.9932
2623 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2624 693/693 [==============================] - 199s
     286ms/step - loss: 0.0233 - accuracy: 0.9931 -
     val_loss: 0.3100 - val_accuracy: 0.9180 - lr: 1.
     0000e-05
2625 Epoch 499/500
2626 692/693 [============================>.] - ETA: 0s
      - loss: 0.0213 - accuracy: 0.9940
2627 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2628 693/693 [==============================] - 199s
```

```
2628 286ms/step - loss: 0.0218 - accuracy: 0.9939 -
     val_loss: 0.3103 - val_accuracy: 0.9201 - lr: 1.
     0000e-05
2629 Epoch 500/500
2630 692/693 [============================>.] - ETA: 0s
      - loss: 0.0259 - accuracy: 0.9922
2631 [Plateau Check] Variação últimas 30 épocas: 0.
     002891
2632 693/693 [==============================] - 198s
     285ms/step - loss: 0.0259 - accuracy: 0.9922 -
     val_loss: 0.3111 - val_accuracy: 0.9194 - lr: 1.
     0000e-05
2633 >> PIPELINE DE TREINAMENTO FINALIZADO COM SUCESSO
2634
2635 Process finished with exit code 0
2636
```